



Information transmission and inefficient lobbying



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ABSTRACT

In a seminal paper, Grossman and Helpman (1994) introduced a framework to understand how lobbying influences the choice of import/export tariffs. In this paper we extend their analysis and assume that lobbies have private information to analyze the effects of information transmission in a Grossman and Helpman lobby game. Information transmission leads to two agency costs. One refers to the cost of signaling and the other to the cost of screening the lobby's competitiveness. The signaling cost reduces distortions resulting from the screening cost. On the other hand, distortions from the screening cost enhance the signaling cost. The two combined effects generate more distortions in the political game than each effect separately. Tariff protection and lobbies' rents are lower than those found in Grossman and Helpman's truthful equilibrium. Therefore, information transmission may improve welfare through the reduction of rent-seeking activities.

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1. Introduction

Lobbying plays a central role in the policy-making process in many fields of economic literature such as trade, taxation and regulation. Yet, there is no consensus about the role of lobbies in the political process. One branch of the literature² treats lobbies as groups that have privileged access to information relevant to the decision-making process, but conflict of interest may however prevent the revelation of this information.

Another branch views lobbies as rent-seeking groups that exercise influence through money contributions. Lobbies swing the decision of an influenceable policy maker in their favor at the expense of the society's welfare. Within this literature, Grossman and Helpman (1994) – GH hereinafter – is the seminal paper to capture the importance of lobbying. In their model, the political game takes place in a small economy and lobbies represent productive sectors that offer money contributions to the policy maker in order to receive tariff protection. A fraction of individuals in the economy is not represented by lobbies and does not participate in the political game. The country's trade policy favors the sectors that lobby while the deadweight loss from tariff protection is borne by individuals who do not lobby. GH assume perfect information in the political game.

This paper stands in-between the two branches. We generalize the GH rent-seeking lobbying model by introducing asymmetric information on the competitiveness of productive sectors. There are two lobbies, each one informed only about

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² For example, Potters and Van Winden (1992), Austen-Smith (1995), Austen-Smith and Wright (1992), Krishna and Morgan (2001), and Esteban and Ray (2006).

its own competitiveness. The policy maker has no private information and does not observe the sectors' competitiveness. Lobbies are rent seekers and have more information regarding the impact of policy instruments.

The political game unfolds as follows. Lobbies simultaneously offer contributions to the policy maker in exchange for tariff protection. The policy maker may accept contributions and grant tariff protection according to the importance he³ gives to the society's welfare. This common agency game can be analyzed as two bilateral relationships between each lobby and the policy maker. The information asymmetry has two distinct effects in these bilateral relationships. A lobby's bilateral relationship with the policy maker is affected by the lobby's private information because the implications of the policy maker's decision depend on it. A signaling problem then arises from this information asymmetry. The two bilateral relationships also affect each other. The tariff protection negotiated in one relationship has impacts on the utility the policy maker obtains in the other bilateral relationship. As a result, the private information of one lobby is important for the rival lobby's bilateral relationship with the policy maker. As we shall see in more detail, this information transmission problem has the structure of a screening problem.

The first asymmetric information problem arises because the policy maker does not know the lobbies' types. Facing the same tariff protection, more competitive lobbies ("high-types") cause greater deadweight loss than less competitive ones ("low-types"). A costly signaling problem is introduced in the lobby/policy maker relationship. Low-type lobbies must separate themselves from high-type lobbies. Separation allows the policy maker to correctly discern lobbies' types, although low-type policies are distorted. We refer to these distortions as the signaling effect (or cost).

The second asymmetric information problem comes from the fact that goods are substitutes and each lobby does not know the rival's type. Substitutability implies that the society's welfare and the policy maker's preference are not separable functions of the lobbies' policies. The marginal deadweight loss from tariff protection in the market of lobby 1 is a function of the tariff protection in market 2 (and vice versa). When lobby 1 does not observe the tariff protection in market 2, she does not access her true marginal cost of tariff protection because it depends on the private information of lobby 2.

In case contributions are separating, the policy maker endogenously learns the lobbies' types. Lobby 1 is then able to make conditional contributions to screen the rival lobby's information through the policy maker. Screening is costly and generates distortions in the political game. We refer to these distortions as the screening effect (or cost). Both asymmetric information problems constitute what we call the information transmission problem.

The distortions resulting from information transmission reduce the lobbies' influence. The low-type lobby separates herself from the high-type lobby by demanding less tariff protection than she would under perfect information. Screening the rival's type makes the lobby leave informational rents to the policy maker and also demand less tariff protection. Therefore, information transmission allows the policy maker to extract informational rents from the lobbies. We show that in such games, tariff protection decreases, imports increase, and the society's welfare increases when compared with the GH's truthful equilibrium.

Not only the pattern of tariff protection but also the distribution of the political game's surplus significantly differ from those in GH. In their paper, when lobbies are highly concentrated (the case we are considering here) the policy maker gets his reserve utility and lobbies extract all the surplus. With privately informed lobbies, the policy maker has bargaining power against the lobbies, due to the screening effect, and is able to retain some rents. Since tariff protection is also different from the truthful equilibrium, some of the political rents are dissipated. Both effects reduce the lobbies' rents.

Our framework also allows us to evaluate the role of transparency in lobbying activity. Some papers which study the role of information in lobby games focus on issues related to costly signaling or cheap talk.⁴ While our paper also features costly signaling, the screening effect is novel in the lobby literature. The screening effect arises because the information transmitted in one bilateral relationship provides the policy maker extra bargaining power to use in his other bilateral relationship. This result raises doubts about the role of the often plagued transparency in the relationship between lobbies and government (policy maker). Making contributions public in our model would decrease the society's welfare because the policy maker would be stripped of an instrument to bargain with lobbies.

This paper also provides a theoretical contribution to the literature. To the best of our knowledge, this is the first model that brings together signaling and screening effects in a common agency game. We are also able to identify how the two information transmission problems interact with each other. We find that increasing the distortions from the screening effect has a significant impact on the separation effort by amplifying the distortions from the signaling effect. More screening distortions lead to more signaling distortions. On the other hand, the signaling effect decreases rents of the political game, which in turn, lowers the informational rent and the distortions from the screening effect. More signaling distortions lead to less screening distortions. The interaction of both information problems generates more distortions in the political game than each problem separately.

In the following subsection we discuss the related literature. Next section presents the economy and the political game, and characterizes the efficient policies as well as the equilibrium of the political game under perfect information. In Section 3, we present the informed lobby problem. We define, prove the existence and characterize the equilibrium of the political game in Section 4. We then compare the equilibrium of the political game with the equilibrium of the game under perfect information. Section 5 provides a discussion of our selected equilibrium. Section 6 concludes. All the proofs are pre-

³ We will use feminine pronouns to refer to lobbies and masculine pronouns in referring to the policy maker.

⁴ See Potters and Van Winden (1992) and Krishna and Morgan (2001).

sented in [Appendix A](#). The online [Appendix B](#) presents the non-negativeness of the equilibrium contributions, countervailing incentives and the interim efficiency of the equilibrium policies.

1.1. Related literature

Lobby games are also modeled as common agency games in [Le Breton and Salanié \(2003\)](#), [Martimort and Semenov \(2008\)](#), and [Campante and Ferreira \(2007\)](#). The first two papers consider the ideological uncertainty case, i.e., the policy maker's preference regarding both contributions and welfare is his private information. [Le Breton and Salanié \(2003\)](#) present a model where the lobby formation is endogenous, which we do not consider here (as in GH, lobbies already exist). [Campante and Ferreira \(2007\)](#) exploit the inequality between lobbies and the presence of credit constraints to show that poorer lobbies exert relatively more influence depending on differences in their opportunity costs.

Our results are more comparable to [Martimort and Semenov \(2008\)](#). They show that ideological uncertainty reduces lobbies' influence making the equilibrium of the political game closer to the policy maker's preferred policies. As political uncertainty grows, the equilibrium eventually becomes free from lobbies's influence. In our case, the screening effect also reduces lobbies' influence and makes the equilibrium policies closer to the policy maker's "preferred policy" (free trade policy). However, the papers have important differences because the lobbies, not the policy maker, hold the private information. In our model, for example, there is no free from influence equilibrium. More importantly, in [Martimort and Semenov \(2008\)](#) there is no signaling effect and, consequently, they cannot analyze the interaction between the screening and signaling effects.

[Ludema et al. \(2010\)](#) and [Tovar \(2011\)](#) are in line with our basic motivation. [Ludema et al. \(2010\)](#) provide empirical evidence on the relevance of information transmission within lobbying activities. Using firm-level data on lobbying contributions, they show that firms manage to influence the probability of securing a smaller import tax for a foreign input (or preventing an import tax reduction for a competing foreign input) using both monetary contributions and verbal messages to the government. The impact of each instrument for government decision is positive, but the impact of verbal messages is about three times larger than contributions. This evidence suggests that both money and especially information transmission are relevant for lobbying. [Tovar \(2011\)](#) estimates the impact of lobbying expenditures and campaign contributions on the pattern of trade policies. This paper relates lobbying expenditure to the informational role of lobbies and finds that lobbying expenditure has a significant impact on the tariff protection received by the sectors.

Within the literature that focuses on the informative role of lobbies, our paper is also related to [Esteban and Ray \(2006\)](#). In their model, lobbies also represent producers that signal their productivity to the policy maker through contributions. Their paper, however, differs from ours in two key aspects. First, they assume that the policy maker is not influenceable. He only tries to allocate resources to more efficient producers. Second, the efficient policy depends on the lobbies' private information. In such case, the policy maker fails allocate resources efficiently because of a signal jamming problem that prevents him from learning the sectors' true productivity. In our model, the policy maker does not need the lobbies' information to select the welfare efficient policy. Inefficiencies in the political game result from the combination of the rent-seeking nature of the lobby activity and information transmission. The information asymmetry reduces the rent-seeking activity and then enhances the welfare. The structures of the two models are very different and so are the qualitative implications of information asymmetries.

[Bennedsen and Feldmann \(2006\)](#) are also related to our paper. They analyzed a common agency game where lobbies search for information about the state of the economy and make contributions to influence the policy maker's decision. The ability to make contributions reduces the lobby's willingness to search for information and competition between lobbies favors those who abstain from searching. Their structure differs from ours because the information gathered does not affect the lobbies' preferences while in our model the information directly affects preferences.

Finally, our approach benefits from [Martimort and Moreira \(2010\)](#) who analyzed a divisible public good provision problem as a common agency game with privately informed contributors. Contributors are privately informed about their willingness to pay for the public good and give conditional money transfers to a common agent who produces the public good. Similarly, we introduce private information into the lobbies' preferences and analyze a common agency game with privately informed lobbies. The screening effect is similar to that studied in their paper. In contrast with their paper, however, our model features common values. The policy maker also cares about the social welfare which includes the lobbies' payoff. [Maskin and Tirole \(1992\)](#) showed that common value element introduces distortions in informed lobby problems in the same spirit of signaling games (e.g., [Spence, 1973](#)). The nature of the signaling effect is then directly related to the common value aspect of our model.

2. Model

The basic framework is similar to the GH model. We consider a small, competitive economy that faces fixed international prices (p^e). The economy has a size one population of consumers. These consumers have preferences for three goods (x^0 , x^1 and x^2) represented by the following utility function:

$$u(x^0, x^1, x^2) = x^0 + \sum_n (\omega - \eta x^n) x^n + \sigma x^1 x^2,$$

where the superscript 0 denotes the numeraire and $n \in \{1, 2\}$ refers to the productive sector n ; σ is a constant, ω and η are positive constants.

The government's revenue from import tax is given by

$$TR = \sum_n (p^n - p^e)(x^n - y^n),$$

where $p^n - p^e$ is the import tariff of good n and the international price p^e is the same for x^1 and x^2 . The home production of good n is y^n . This revenue is redistributed to the society through lump-sum transfers.

Good 0 is not taxed and its international price is normalized to 1. It is produced only from labor with constant returns of scale with an input-output coefficient of 1. We assume that the labor supply is sufficiently large that wages can be also normalized to 1.

The wages and the government transfers define the consumers' income which, together with preferences, allow us to find the market demands:

$$x^n = \max\{a - bp^n + dp^{-n}, 0\},$$

where $b > 0$ and d is a parameter that defines whether goods are substitutes ($d > 0$) or complements ($d < 0$). In practice, we consider a large enough such that the market demand for home goods will not be zero on the relevant price range.⁵

Assumption 1. Goods x^1 and x^2 are substitutes ($d > 0$).

We focus on $d > 0$ for convenience but most of the results remains true for complementary goods. When d is not zero, demands are interdependent and the policies in one sector affects the deadweight loss from policies in the other sector. This is an important difference between our model and that of GH (where $d = 0$) and is a key assumption for the information transmission problem in this model.⁶

Substituting the demands into the utility function we can compute the indirect utility function, denoted by $u(p^1, p^2)$, with some abuse of notation.

Goods are produced with sector specific productive factors. Owners of these factors receive all the profit from production. We assume that owners of productive factors are a negligible fraction of the population so that the factor ownership is highly concentrated. We refer to the owners of the specific factor as producers.

The production technology of goods x^1 and x^2 is represented by a positive and increasing marginal cost function with capacity constraint:

$$\frac{\partial c}{\partial y}(\theta, y) = \begin{cases} \frac{\gamma y}{\theta} & \text{if } y \leq \frac{\theta}{1-\gamma} \\ \infty & \text{if } y > \frac{\theta}{1-\gamma}. \end{cases}$$

Notice that the elasticity of the supply function will be different depending on whether the optimal production is an interior or a corner solution. In the first case, production increases with home prices, while in the second, production is fixed. For simplicity, we only analyze two polar cases. In the first case ($\gamma = 0$), each sector produces exactly the capacity constraint. In the second case ($\gamma = 1$), the capacity is never reached.⁷ As we shall see, these extreme cases result in different information transmission problems.

The profit function is given by $\theta\pi(p)$, where $\pi(\cdot)$ is a convex function depending on the value of γ . The supply function of good n is denoted by $y^n(\theta^n, p^n)$. By the envelope theorem, $y^n(\theta^n, p^n) = \theta^n \pi'(p^n)$.

The welfare is the sum of the government's revenues and the consumers' as well as producers' surpluses in all markets:

$$W(\theta^1, p^1, \theta^2, p^2) = u(p^1, p^2) + \sum_n (p^n - p^e)(x^n(p^n, p^{-n}) - \theta^n \pi'(p^n)) + \sum_n \theta^n \pi(p^n).$$

Fig. 1 presents the deadweight loss from a tariff protection increase in market (of good) 1.

In market 1 the home price $\bar{p}^1$ is above the international price p^e due to the tariff protection. The downward sloping line is the home market demand and the upward sloping line is the home supply of good 1. The triangle A below the demand curve and above the home price is the consumers' surplus; B and F are the producers' surplus; D is the tariff revenue; C and E are the deadweight loss of the tariff. The rectangle G in market 2 is an extra revenue due to substitutability and the increase of tariff protection in market 1.

The impact of tariff protection on welfare is given by

$$\frac{\partial W}{\partial p^n} = -(b + \theta^n \pi''(p^n))(p^n - p^e) + d(p^{-n} - p^e). \quad (1)$$

⁵ That is, equilibrium prices will never reach $\frac{a}{(b-d)}$ when a is large enough.

⁶ The only exception is Proposition 2 where we consider the case $d = 0$.

⁷ The results that follow are essentially the same for $\gamma > 0$ once production does not reach the capacity constraint.

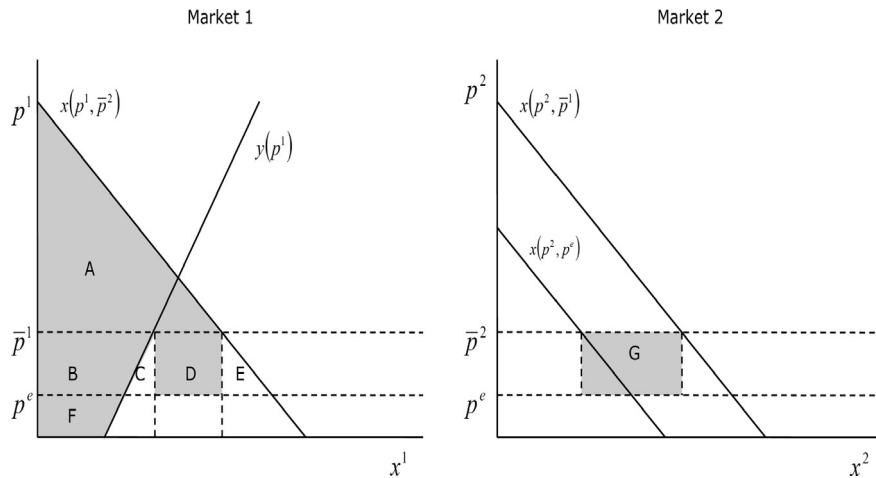


Fig. 1. The welfare impact of a tariff increase.

Notice that the area of triangle E is $b(p^1 - p^e)$ in (1) and represents the deadweight loss from the decrease in home consumption. The area C is the term $\theta^1 \pi''(p^1)(p^1 - p^e)$ in (1) and represents the deadweight loss due to import substitution. Finally, the area of rectangle G is the last term in (1). Eq. (1) shows that welfare is decreasing with respect to the sector's tariff protection. Also, the marginal deadweight loss is bigger as the sector's protection increases and is smaller as the other sector's protection increases, since d is positive. This derivative plays an important role for the strategic interactions of the political game. The term $\theta^n \pi''(p^n)(p^n - p^e)$ is related to the signaling effect since the lobby's type affects directly the marginal deadweight loss from tariff protection. On the other hand, the term $d(p^{-n} - p^e)$ is directly related to the screening effect, because the marginal deadweight loss from protection of sector n depends on the protection of sector $-n$ which, in equilibrium, depends on the rival lobby's type and relates the influence of one lobby with the cost of influence of the other.

The government's revenue, the market demands, the home supplies, and the international prices define the economy in our model.

2.1. Political game

There are three players: two lobbies and one policy maker. Lobbies offer contributions, $C \in \mathbb{R}_+$, to the policy maker. They are the principals of the common agency game. We assume that consumers cannot lobby.

Each lobby represents the producers of her sector. They care about the profit of the sector they represent and dislike giving contributions to the policy maker. Their utility function is

$$V(\theta, p, C) = \theta \pi(p) - C.$$

Since the sectors' ownership is highly concentrated, lobbies do not care about the consumer's surplus.

The policy maker is the common agent who chooses the home prices of the economy, $p^1, p^2 \in \mathbb{R}_+$, by imposing an import tariff.⁸ Home prices are the only economic policies available to him. He cares about the social welfare (W) but also likes contributions, and is willing to trade economic welfare for contributions. His preferences are represented by

$$U(\theta^1, p^1, C^1, \theta^2, p^2, C^2) = \sum_n C^n + \lambda W(\theta^1, p^1, \theta^2, p^2),$$

where $\lambda > 0$ is the weight of welfare.

2.1.1. Asymmetric information

The competitiveness parameter θ can take two values: θ_l and θ_h , where $\theta_h > \theta_l$. The distribution of θ 's is common knowledge, i.i.d. and z is the probability of $[\theta = \theta_h]$. Its realization is private information of the lobby, so each lobby knows her type but does not know her rival's type, and the policy maker does not know their types.

We make the following assumption about the parameters to assure interior solutions of the lobbies' problems:⁹

⁸ Using import tariff, the policy maker cannot set prices above the competitive price of the closed economy. We ignore this upper bound by assuming that the policy maker can collect lump-sum taxes to pay an export subsidy. Hence, the home price can be above the competitive price of the closed economy.

⁹ This assumption is sufficient to obtain interior solutions and greatly simplifies our analysis since it rules out negative prices. Essentially, it ensures an interior solution for the "virtual utility" maximization problem.

Assumption 2.

$$(1 - z)b > d,$$

$$2\theta_l \geq \theta_h,$$

and if $\gamma = 1$,

$$(1 - z)(b + (1 - \lambda^{-1})\theta_h) > d.$$

The first inequality in [Assumption 2](#) states that the substitutability between goods is not very large. The second inequality states that the difference between the asymmetric information parameters is moderate. The third has a similar role as the first one when $\gamma = 1$.

2.1.2. Strategy space

A contribution schedule of lobby n , $C^n(\theta^n, p^n)$, specifies the level of contribution $C^n \in \mathbb{R}_+$ for each policy $p^n \in \mathbb{R}_+$ and type $\theta^n \in \{\theta_l, \theta_h\}$. We assume, for sake of simplicity and reality, that a lobby cannot make contributions conditional on the rival's policy p^{-n} .

Once the contribution schedule is accepted, the policies are implemented and payments are made accordingly (we then are assuming commitment in the political game).

Since the political game is symmetric, we drop the superscript index whenever this does not cause any further confusion.

2.1.3. Timing

1. Nature draws the lobbies' types and each lobby learns her type;
2. Each lobby non-cooperatively offers contribution schedules to the policy maker;
3. The policy maker either accepts or rejects contributions; and,
4. The policy maker chooses policies and, when contributions are accepted, payments are made accordingly.

The preferences of the lobbies and the policy maker, the information structure, the strategy space, and the timing define the political game.

This rent seeking model has two benchmarks that will help us to evaluate the effects of information transmission. The first one is the free trade equilibrium. It defines which policies arise if there is no political influence on the decision-making. The second benchmark is the truthful equilibrium. It defines which policies arise from the political game if there is no information asymmetry.

2.2. Free trade

If the policy maker does not care about contributions, he chooses policies that maximize society's welfare:

Definition 1. The free trade equilibrium policies $(\hat{p}_{ik}, \hat{p}_{ki})$ are defined by

$$(\hat{p}_{ik}, \hat{p}_{ki}) \in \arg \max_{p_{ik}, p_{ki}} W(\theta_i, p_{ik}, \theta_k, p_{ki}),$$

where the first subscript index refers to the lobby's own type, the second index refers to the rival's type, and where $i, k \in \{l, h\}$.

The first-order conditions of this problem are given by

$$\frac{\partial W}{\partial p^n}(\theta_i, \hat{p}_{ik}, \theta_k, \hat{p}_{ki}) = 0, \quad (2)$$

for all n and k such that $k \neq i$.

From (1) we obtain that the free trade equilibrium is the welfare maximum for the society,¹⁰ i.e., $\hat{p}_{ik} = \hat{p}_{ki} = p^e$. Any deviation from these policies reduces welfare.

2.3. Truthful equilibrium

From [Bernheim and Whinston \(1986a, 1986b\)](#) we know that when principals play truthful strategies,¹¹ the solution of the common agency game is the same as the solution of a centralized problem that maximizes the surplus of the political game.

¹⁰ In this framework, the second-order condition implies that free trade is the welfare maximum whenever $b > d$, and trivially follows from [Assumption 2](#).

¹¹ Given the model's primitives, truthful strategies are characterized by contribution schedules such that $\partial C / \partial p = \theta \pi'(p)$.

Definition 2. The truthful equilibrium policies $(\bar{p}_{ik}, \bar{p}_{ki})$ are defined by

$$(\bar{p}_{ik}, \bar{p}_{ki}) \in \arg \max_{p_{ik}, p_{ki}} \theta_i \pi(p_{ik}) + \theta_k \pi(p_{ki}) + \lambda W(\theta_i, p_{ik}, \theta_k, p_{ki}).$$

The first-order conditions resulting from the truthful contribution schedules are

$$\theta_i \pi'(\bar{p}_{ik}) - \lambda \frac{\partial W}{\partial p^1}(\theta_i, \bar{p}_{ik}, \theta_k, \bar{p}_{ki}) = 0, \quad (3)$$

and a symmetric first-order condition for $\bar{p}_{ki}$.

These first-order conditions equalize the marginal benefit of the lobbies and the marginal deadweight loss for the society. Compared to condition (2), (3) gives an extra weight to lobbies, and therefore policies increase for lobbies and the deadweight loss is borne by the rest of society.

For $\gamma = 0$, policies implemented by the truthful equilibrium are given by

$$\bar{p}_{ik} = \frac{\theta_i b + \theta_k d}{\lambda(b^2 - d^2)} + p^e, \quad (4)$$

and for $\gamma = 1$, policies are given by

$$\bar{p}_{ik} = \left(\frac{\theta_i(\lambda(b + \theta_k) - \theta_k) + \lambda d \theta_k}{(\lambda(b + \theta_k) - \theta_k)(\lambda(b + \theta_i) - \theta_i) - \lambda^2 d^2} + 1 \right) p^e. \quad (5)$$

The free trade equilibrium defines the efficient solution and the truthful equilibrium defines the solution of the political game without asymmetric information. We will keep these two benchmarks in mind as we compare the qualitative properties of the political game with privately informed lobbies.

3. The informed lobby problem

In this section, we present the informed lobby's best-response problem. In the political game, lobbies simultaneously offer contribution schedules to the policy maker. Each lobby is privately informed about her type and does not know her rival's type. Therefore, the lobby's utility maximization problem can be tackled as an informed principal problem with the policy maker. Some remarks are in order. In this informed principal problem, each lobby has private information about her type while the policy maker has no a priori private information and simultaneously receives contributions from both lobbies. When these contributions are separating, the policy maker learns the lobbies' types before implementing the policies.¹²

From Maskin and Tirole (1992), we know that an informed principal does not gain by postponing information revelation to the agent when both principal and agent have quasi-linear preferences. The political game with informed lobbies becomes, from the perspective of each lobby, a principal-agent problem where both parties have private information: the lobby on her type and the policy maker on the rival's type.

The strategy to solve the informed lobby problem is adapted from Martimort and Moreira (2010), who provide a solution to a common agency game with informed principals. Their approach cannot be directly applied to our model, however, because of differences between the two setups.¹³

We now present our approach in more detail. To solve the lobby's best-response problem we must describe the conjectures each lobby has about her rival's contribution schedules. The set of best responses to all possible conjectures the lobby may have about her rival's contribution schedule is potentially very large. We opt to restrict the lobby's conjecture about the rival's contributions to the set of separating contributions. Consequently, the set of possible strategies that can emerge as solutions is reduced and as well as the set of equilibria we will characterize later on. This restriction of strategies leads us to characterize fully revealing equilibria. We formally present the restriction in the selection criterion in what follows.

From now on let us omit the superscript n and write the rival's variables in bold. The rival's contribution schedule is denoted by $\mathbf{C}(\theta_k, \mathbf{p})$ and the policy by $\mathbf{p}_{ki}$, where the subscript indexes i and k are reversed when compared to the indexes of the type- i lobby's variables.

Selection Criterion (SC). The rival's contribution schedule is separating ($\mathbf{C}(\theta_h, \mathbf{p}) > \mathbf{C}(\theta_l, \mathbf{p})$) and the policy is strictly increasing in her type ($\mathbf{p}_{hi} > \mathbf{p}_{li}$).

¹² Remember that lobbies' offers are simultaneous. We can interpret that the contribution schedule offer is part of the negotiation protocol between lobbies and the policy maker. The policy maker may learn lobbies' private information and they cannot observe her rival contribution schedule (i.e., negotiation protocol). Therefore, although policies are ex-post observable, the assumption that lobbies cannot condition on the rival policies is without loss.

¹³ Martimort and Moreira's approach was designed for a common agency game with informed principals, but with continuous types, a public good policy, and private value preferences for the agent. Our setup, on the other hand, features discrete types, a single policy for each principal, and common value preferences for the agent.

Restricting our attention to contributions and policies that are strictly increasing in type is intuitive since high-type lobbies have higher valuation for tariff protection. Under the SC a lobby expects her rival to offer separating contributions, which in turn, implies that the policy maker learns the rival lobby's type directly from her offer. Hence, it allows us to tackle the informed lobby problem as if the policy maker holds the rival's private information. To compute the equilibrium, we assume that the SC holds for the rival's policy and solve the lobby's best-response problem. We must then check that the SC is satisfied for the lobby's best-response policy. In the online [Appendix B](#) we present a formal justification for the SC.

Focusing on separating equilibria requires incentive compatibility constraints on the lobby side. Let $C(\theta_i, p)$ be a contribution schedule and p_{ik} be a policy choice of type- i lobby. To simplify notation, we will denote the contribution schedule $C(\theta_i, p_{ik})$ by C_{ik} . The lobby's incentive compatibility constraint states that a type- $(-i)$ lobby does not want to behave as type- i lobby:

$$E[\theta_{-i}\pi(p_{-i.}) - C_{-i.}] \geq E[\theta_{-i}\pi(p_{i.}) - C_{i.}], \quad (IC_{-i})$$

where $-i \neq i$ is the other type the lobby could assume. This constraint ensures that the policy maker can correctly learn the lobby's type from the contribution schedule. When binding, these constraints generate distortions in the same fashion as in signaling games. These distortions constitute what we call the signaling effect (or cost).

The lobby's individual rationality constraint

$$E[\theta_i\pi(p_{i.}) - C_{i.}] \geq \theta_i\pi(p^e) \quad (IR_i)$$

ensures that her expected utility is at least the utility of the free trade policy with no contribution.

The policy maker learns the rival's type when he receives her offer (before implementing policies) as the rival's offer is separating. In turn, the rival's offer is valuable information for the type- i lobby because the marginal cost of tariff protection is decreasing with respect to the rival's policies. The lobby may want to screen the policy maker in order to learn the rival's information. Screening requires incentive compatibility constraints for the policy maker. These incentive compatibility constraints ensure that he chooses policy levels according to the true type of the rival lobby. In other words, they ensure that the policy maker chooses the contribution associated with her true cost:

$$C_{ik} + C_{ki} + \lambda W(\theta_i, p_{ik}, \theta_k, p_{ki}) \geq C_{i(-k)} + C_{ki} + \lambda W(\theta_i, p_{i(-k)}, \theta_k, p_{ki}), \quad (ICP_{ik})$$

where $-k \neq k$ is the rival's fake type.

Notice that constraint (ICP_{ik}) leads to two different constraints in the type- i informed lobby problem (one for each possible k). The distortions coming from these constraints constitute what we call the screening effect (or cost).

Finally, the policy maker's individual rationality constraint

$$C_{ik} + C_{ki} + \lambda W(\theta_i, p_{ik}, \theta_k, p_{ki}) \geq \lambda W(\theta_i, p^e, \theta_k, p^e) \quad (IRP_{ik})$$

ensures that the policy maker is prompted to accept contributions that give him at least his reserve utility.¹⁴

3.1. Type- i informed lobby problem

Let us fix type- i lobby and her rival's contribution schedules $C(\theta_k, p)$ and policies p_{ki} satisfying the SC throughout this subsection, where $k = l, h$. Now we can present the type- i informed lobby problem:

$$\max_{\substack{p_{ih}, p_{il} \\ C_{ih}, C_{il}}} E[\theta_i\pi(p_{i.}) - C_{i.}] \quad (6)$$

subject to (IC_{-i}) , (IR_i) , (ICP_{ik}) , (IRP_{ik}) , and $C_{ik} \geq 0$ for all $-i$ and k .

Notice that the expectation in the objective function of problem (6) is conditional on the realization of type θ_i and, therefore, taken with respect to the rival's type. To obtain the complete characterization of the solution of problem (6) we must solve it for every realization of lobbies' types. We begin identifying which constraints are binding at the optimal solution. Because goods are substitutes ([Assumption 1](#)), the cross derivative of the welfare function is $d > 0$ and then the marginal deadweight loss from a policy (1) is inversely related to the size of the policy in the other market. Policies are then strategic complements for the lobbies' problems. A lobby prefers to face a high-type opponent because the marginal deadweight loss from the policy (1) is smaller than the one when facing a low-type opponent.

Lemma 1. Suppose that constraints (ICP_{ih}) and (IRP_{il}) hold. Then, the type- i lobby's policy is increasing in the rival's type:

$$p_{ih} \geq p_{il}. \quad (7)$$

¹⁴ More precisely, the individual rationality constraint should give the policy maker the utility he receives by rejecting the contribution of one lobby and accepting the rival's. However, with substitute goods ([Assumption 1](#)), this is exactly the utility he gets from rejecting both contributions.

Lemma 1 suggests which of the policy maker's incentive compatibility constraints should bind. In the absence of proper incentives, the policy maker will be prompted to lie and choose the policy as if the rival is the low type (i.e., high deadweight loss state) when she is truly the high type. If constraints (IC_{ih}) and (IR_{il}) are binding, we can eliminate the contribution variables C_{ik} from problem (6). We then solve it with respect to the policy variables p_{ik} as in the tradition of the contract theory literature. Constraint (IC_{-i}) may or may not be binding, however, depending on the parameters of the model. Moreover, we assume that constraint (IR_i) is satisfied and we must check it ex-post facto.

Lemma 2. Suppose that constraints (IR_{il}) and (IC_{ih}) are binding. Then the first-order conditions of the type- i informed lobby problem are given by

$$\theta_i \pi'(p_{ik}) - \lambda \frac{\partial W}{\partial p^1}(\theta_i, p_{ik}, \theta_k, p_{ki}) + \frac{\mu_{-i}}{1 - \mu_{-i}} \Delta \theta_i \pi'(p_{ik}) + \lambda I(k) \frac{z}{1 - z} d(p_{li} - p_{hi}) = 0, \quad (8)$$

along with the usual slackness conditions for the constraint (IC_{-i}) , where μ_{-i} is the Lagrangian multiplier of (IC_{-i}) , $\Delta \theta_i = \theta_i - \theta_{-i}$, and

$$I(k) = \begin{cases} 1, & \text{if } k = l \\ 0, & \text{if } k = h. \end{cases}$$

The first two terms of (8) are the lobby's marginal benefit and the deadweight loss from the policy, which are the driving forces that determine the truthful equilibrium policies (see Eq. (3)). The third term captures the cost of separation, where μ_{-i} is the shadow price of constraint (IC_{-i}) . Notice that $\Delta \theta_i < 0$ and the low-type has to demand a lower policy to separate herself.

The last term of (8) is due to the informational rent the lobby has to leave to the policy maker in order to induce him to tell the truth. She demands a lower policy when facing a low-type rival to save on informational rents, which decreases her utility as well. This rationale constitutes the usual trade-off between policy efficiency and rent extraction.

Constraint (IC_{-i}) may be binding depending on the shape of the profit function. We now divide the analysis into two polar cases according to the value of the parameter γ . The following lemma greatly simplifies the analysis.

Lemma 3. The type- i informed lobby problem is such that

- (i) when $\gamma = 0$, constraint (IC_{-i}) is never binding for all $-i$; and,
- (ii) when $\gamma = 1$, constraint (IC_l) is never binding.

When $\gamma = 0$, the derivative of the supply function with respect to the policy is zero and, consequently, the marginal deadweight loss from the policy (1) is not a function of the lobby's type. Therefore, the lobby's information is not directly relevant for the policy maker and the signaling constraint (IC_{-i}) is not binding (Lemma 3(i)). When $\gamma = 1$, the derivative of the supply function with respect to the policy is θ and the marginal deadweight loss from the policy (1) is an increasing function of the type. In this last case, the signaling effect may arise. Yet low-type lobbies will never mimic high-type lobbies once high-types have to give higher compensations to the policy maker (once the deadweight loss is increasing in type). Therefore, constraint (IC_l) is never binding (Lemma 3(ii)), and we conclude that constraint (IC_{-i}) may only bind in the low-type informed lobby problem.

3.2. Interactions between screening and signaling

The approach described in Section 3.1 tackles each lobby's utility maximization problem as an informed principal problem. Lobbies and the policy maker have private information, introducing both signaling and screening into this principal-agent relationship. To the best of our knowledge, this is a novel contribution to the literature and allows us to analyze how signaling and screening interact in an informed principal problem.

To understand the interaction between signaling and screening, we must be able to evaluate the consequences from enhancing the intensity of the signaling effect over the screening effect and vice-versa. The natural parameter to capture these intensities would be $\Delta \theta_h$ which measures the private information stake for the informed lobby problem. However, if we keep the symmetry of the model, $\Delta \theta_h$ impacts on both signaling and screening effects. We instead look for other parameters that only affect each information transmission problem separately and keep the model's symmetry.

A natural parameter related to the screening effect is the probability of the high type z .¹⁵ This parameter directly impacts on the hazard rate term of Eq. (8). Higher z implies higher distortion from the screening effect. To find an independent parameter related to signaling is trickier. In Eq. (8) the shadow price of constraint (IC_{-i}) , μ_{-i} , is directly related to the intensity of the signaling effect. Parameter μ_{-i} is, however, endogenous and does not qualify for the comparative statics

¹⁵ We could also have selected parameter d . But this substitutability parameter increases welfare while keeping profits constant and, consequently, has an impact on the signaling effect that is not related to the screening effect.

exercise. In order to identify a parameter that impacts μ_{-i} , let us take a step back and allow the policy maker to form pessimistic beliefs which can lead to an even higher separation cost. With pessimistic beliefs separation may not be achieved when type-(-i) lobby is indifferent between two policies. Consequently, policies have to be even more distorted to ensure separation. Let $\bar{U}$ be the utility the policy maker has with a pessimistic belief the policy maker has about the type-(-i) lobby's expected utility.

To perform the comparative statics on the signaling effect, we replace (IC_{-i}) by

$$\bar{U} \geq E[\theta_{-i}\pi(p_i) - C_i] \quad (9)$$

in the informed lobby problem, where $\bar{U} < E[\theta_{-i}\pi(p_{-i}) - C_{-i}]$ and μ_{-i} now represents the Lagrangian multiplier of constraint (9). The parameter $\bar{U}$ is inversely related to the intensity of the signaling effect and will be inversely related to μ_{-i} .

The last step is to provide measures related to the intensity of the signaling and screening effects. The measure for the signaling effect, as we mentioned before, is the shadow price of constraint (IC_{-i}) , i.e., μ_{-i} . This multiplier induces the shift in policies due to the signaling incentive constraint, as can be seen from Eq. (8). The measure for the size of the screening effect is not so direct, because this effect is related to both informational rents and policy distortions. The last term on the right side of Eq. (8), however, is directly related to the screening effect. We define the size of this term as the policy distortion due to the screening effect. We can then state the following:

Proposition 1. *Replacing (IC_{-i}) by (9) in the informed lobby problem (6), its solution is such that:*

- (i) *the screening policy distortions and informational rents increase with $\bar{U}$; and,*
- (ii) *the Lagrangian multiplier of constraint (9), μ_{-i} , increases with z .*

Proposition 1(i) states that if beliefs induce higher separation effort (smaller $\bar{U}$), then the screening effect reduces. The intuition is that separation reduces the surplus of the political game and, hence, the informational rent decreases. The distortion due to the screening effect also becomes less stringent once informational rents are already reduced by the need for separation.

Proposition 1(ii) states that higher z results in higher cost to achieve separation, since the shadow cost of constraint (9) increases. The intuition is that higher z increases the rent extraction of the screening effect which has a positive impact on the high-type payoff when choosing the low-type policies. The low-type lobby then has to further decrease her policies to keep constraint (9) satisfied.

The interplay between the two information transmission problems shows that more intense signaling effect reduces the distortions coming from the screening effect, while enhancing the screening effect increases the distortions coming from the signaling effect.

4. Information transmission and tariff protection

In this section, we define, show the existence and characterize the equilibrium of the political game. We also discuss the effects of information transmission on the pattern of tariff protection given to the lobbies. As indicated earlier, we only consider $\gamma \in \{0, 1\}$ since it greatly simplifies the analysis while covering a rich set of profit functions. We begin by defining our equilibrium concept.¹⁶

Definition 3. A (symmetric) equilibrium of the political game is a pair of (identical) contribution schedules that simultaneously solves problem (6) for each possible lobbies' types i and k and that satisfies the SC for each lobby.

The equilibrium is a fixed point of the best responses derived from the informed lobby problem for all possible types. We then have:

Theorem 1. *If $\gamma \in \{0, 1\}$, there exists a symmetric equilibrium of the political game.*

We now move to the characterization of the equilibrium.

4.1. Capacity constraints ($\gamma = 0$): pure screening case

When lobbies have a binding capacity constraint, the home country supply curve is perfectly inelastic and triangle C in Fig. 1 does not exist. There is no import substitution and sectors with different output capacities generate the same deadweight loss for a given level of tariff protection. The welfare cost comes solely from the decrease in consumption

¹⁶ Remember that we are focusing on equilibria that satisfy the SC.

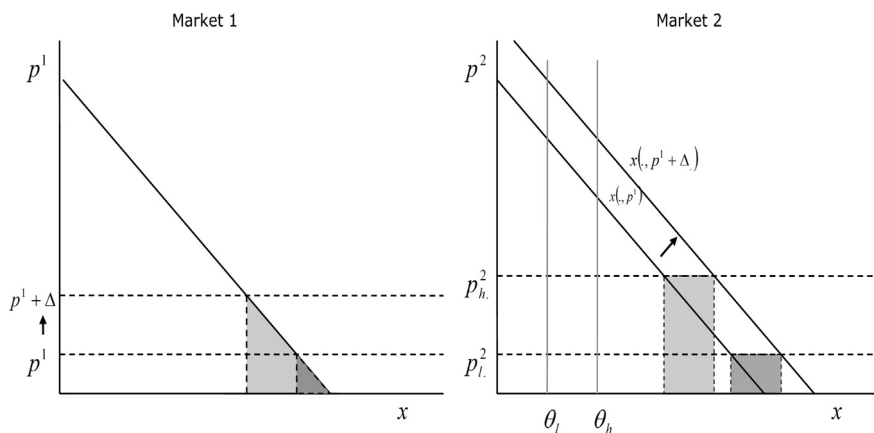


Fig. 2. The screening effect.

(triangle E in Fig. 1) because the term in (1) related to signaling is zero. The marginal deadweight loss does not depend on the lobbies' private information. Only the screening constraints are binding (see Lemma 3(i)).

The tariff in one market affects the demand and, consequently, the marginal cost of the tariff (1) in the other market since demands are interdependent. If a lobby does not know her rival's type, she does not know the true deadweight loss from the tariff protection (1). On the other hand, the policy maker receives contributions that reveal the lobbies' types before implementing the tariff protection. The private information of the rival lobby becomes the "private information" of the policy maker and the lobby has to screen the rival's information through the policy maker.

Fig. 2 presents the effect of a tariff increase in both markets. For simplicity, we assume that $p^e = 0$. Fig. 2 shows a tariff increase from p^1 to $p^1 + \Delta$ and the welfare losses in market 1 for each tariff. The darker triangle is the deadweight loss of p^1 and the lighter triangle is the deadweight loss of $p^1 + \Delta$. A tariff increase in market 1 also shifts positively the demand in market 2 and gives the policy maker an additional tariff revenue equal to the grey rectangles. The lighter rectangle corresponds to the additional revenue when tariff in market 2 is high and the darker rectangle corresponds to the additional revenue when the tariff is low.

In order to get tariff protection, a lobby has to compensate the policy maker for the deadweight loss caused by the tariff increase. The lobby deducts the additional revenue in the other market from the contributions she gives to the policy maker. Lobbies do not know the tariff in the rival market because they do not know the rival lobby's type. Since the policy maker will hold the rival's information, the lobby can only screen this information from him.

This information problem arises because the lobby cannot observe the contribution of the rival lobby once offers are simultaneous. The policy maker learns information that lobbies do not observe, and he can bargain with them. That is, the lobby's inability to observe the rival's contributions gives the policy maker power to extract informational rents. As in traditional screening problems, the lobby distorts policies whenever she faces a low-type opponent in order to save informational rent she has to leave to the policy maker.

Proposition 2. If $\gamma = 0$, the policies of the symmetric and separating equilibrium of the political game are given by:

$$p_{ik}^S = \bar{p}_{ik} - (bI(k) + dI(i))\Psi, \quad \text{for all } i \text{ and } k,$$

where $\bar{p}_{ik}$ is given by (4), $\Psi > 0$ is constant depending on the parameters, $I(h) = 0$ and $I(l) = 1$, and the superscript S refers to screening. Contributions are obtained from the binding constraints (ICP_{ih}) and (IRP_{il}).

Notice that policies decrease except when both lobbies are high-types. The screening effect makes lobbies demand lower policies than they do under perfect information.

One important question is how the screening equilibrium compares with the truthful equilibrium. First, the screening effect gives bargaining power to the policy maker so that lobbies have to pay him informational rents. These informational rents make lobbies distort downward their requested policies when compared to the truthful equilibrium. Since the latter maximizes the political rents, information transmission reduces political rents. Moreover, once policies are above the free trade level in the truthful equilibrium, decreasing them is welfare enhancing. Thus, information transmission increases welfare compared to the perfect information situation.

Corollary 1. The welfare of the screening equilibrium in the political game with informed lobbies is higher than the welfare of the truthful equilibrium in the political game under perfect information.

At this point, we are able to understand the consequences of relaxing Assumption 2. Simple algebra shows that if $d = (1 - z)b$ and $\theta_h = 2\theta_l$, then $p_{il} = p^e$, i.e., free trade policy is achieved when both lobbies are low types. In fact, p^e is a

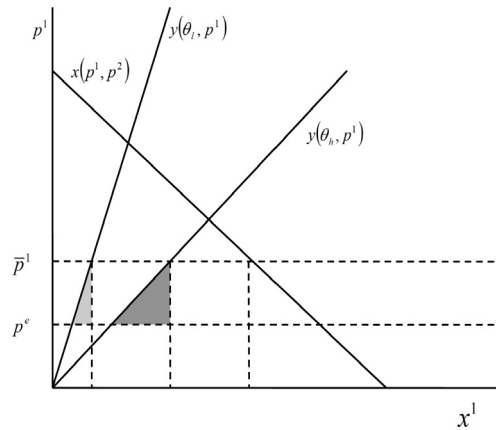


Fig. 3. The signaling effect.

corner solution for the equilibrium policies in the sense that, when this policy is reached, incentives must be provided only through contributions. Assuming that $b > d$, a lobby will always ask for positive tariff protection when she faces the high type ($p_{ih} > p^e$). Therefore, there always exists surplus to be extracted from the lobby/policy maker relationship. Tariff protection is always positive in expected terms, regardless of the policy maker's value of welfare (λ) and of the stake of the information asymmetry ($\Delta\theta_h$).

Proposition 2 contrasts with the result found in [Martimort and Semenov \(2008\)](#). In their common agency lobby game when the policy maker cares a lot about deviations from his policy target and the political uncertainty is large, lobbies refrain from exerting influence. As opposed to this result, our model features positive expected tariff protection. The difference between the results is related to the nature of the information problem. In [Martimort and Semenov \(2008\)](#) both lobbies are eliciting the same information while in our case each lobby is eliciting the rival's information.

4.2. Linear marginal cost ($\gamma = 1$)

When $\gamma = 1$, the profit function is $\theta\pi(p) = \theta p^2/2$ and the home supply is $\theta y(p) = \theta p$, which is more elastic the higher is θ . Different supply elasticities imply different welfare costs of tariff protection (1) as shown in Fig. 3.

Fig. 3 shows that more competitive sectors generate higher welfare costs because they substitute more imports than less competitive ones for a given level of tariff protection. Import substitution is harmful because home consumers buy a good produced at a higher marginal cost than the international price. As high-types cause higher welfare costs, constraint (IC_l) is never binding ([Lemma 3\(ii\)](#)).

Since the policy maker does not know the true value of θ , high-type lobbies may pretend they are low-types in order to give smaller contributions for the policy increase. If both types of lobbies offer the same contribution, the policy maker cannot learn the lobbies' types and has to ask for a compensation that equals the average welfare cost. The low-type lobby has to separate herself to allow the policy maker to learn her type and consequently to pay for the true cost of tariff protection (1), which is smaller than the average cost.

4.2.1. Pure signaling

We begin our analysis by looking at the simpler case of $d = 0$. From Eq. (1) it is easy to see that the term related to screening is zero, and so the policy maker's preferences are separable. When $d = 0$ each lobby does not condition her policy on the rival's type, and hence $p_{ll} = p_{lh}$ and $p_{hh} = p_{hl}$. The high-type policy solves Eq. (8) for $i = h$, and the low-type policy is derived from the lobby's incentive compatibility constraint (IC_h).

Proposition 3. If $d = 0$ and $\gamma = 1$, the policies of the symmetric and separating equilibrium¹⁷ of the political game are given by

$$p_{hh}^* = p_{hl}^* = \frac{\lambda(b + \theta_h)}{\lambda(b + \theta_h) - \theta_h} p^e = \bar{p}_{hh} = \bar{p}_{hl},$$

and also

(i) if $\Delta\theta_h \bar{p}_l^2 \geq \lambda\theta_h(\bar{p}_h - p^e)$, then

$$p_{lh}^* = p_{ll}^* = \frac{\lambda(b + \theta_l)}{\lambda(b + \theta_l) - \theta_l} p^e = \bar{p}_{lh} = \bar{p}_{ll}, \quad \text{and}$$

¹⁷ Notice that we are implicitly choosing the least cost separating equilibrium, i.e., the one that survives the intuitive criterion.

(ii) if $\Delta\theta_h \bar{p}_l^2 < \lambda\theta_h(\bar{p}_h - p^e)$, then

$$p_{lh}^* = p_{ll}^* = \frac{\lambda(b + \theta_l) - \theta_h \sqrt{\frac{\lambda \Delta\theta_h}{\lambda(b + \theta_h) - \theta_h}}}{\lambda(b + \theta_l) - \theta_h} < \bar{p}_{lh} = \bar{p}_{ll},$$

where $\bar{p}_{ik}$ is given in (5) and the superscript $*$ indicates the equilibrium policy. Contributions are computed from the binding constraints (ICP_{ih}) and (IRP_{il}).

Proposition 3 shows that if $\Delta\theta_h/\theta_h$ is relatively large compared to λ , then separation is costless. Otherwise, the low-type lobby has to separate herself by demanding a lower tariff protection. The intuition for this result is the following. The high-type lobby has a higher valuation for tariff protection but also has to give the policy maker a higher compensation for the marginal tariff protection. When λ is relatively small, the marginal cost of tariff protection is small so that the high-type lobby does not pretend she is low-type. When λ is relatively large on the other hand, the deadweight loss from tariff protection is large and the high-type lobby may pretend she is the low-type. Then, the low-type lobby has to ask for a separating policy.

The signaling effect distorts policies in the same direction as the screening effect. Both effects undermine the lobby's influence.

4.2.2. Screening and signaling

If $d > 0$, then both signaling and screening effects coexist. From Section 3.2 we know that these two information transmission problems interact with each other within the informed lobby problem. However, it is extremely difficult to translate these effects into the equilibrium allocation once they are significantly more complex than those within the informed lobby problem. We are not able to fully pin down the closed form solution for the policies due to the complexity of the expressions. Nevertheless, we show that these effects reinforce each other, so the policies are smaller than if we had introduced each effect separately.

Proposition 4. Suppose that $d > 0$ and $\gamma = 1$. If constraint (IC_{-i}) is binding, the policies of the symmetric separating equilibrium of the political game are such that:

$$p_{hh}^* = \bar{p}_{hh}, \quad p_{hl}^* < \bar{p}_{hl}, \quad p_{lh}^* < \bar{p}_{lh}, \quad \text{and} \quad p_{ll}^* < \bar{p}_{ll},$$

where $\bar{p}_{ik}$ is given in (5). Contributions are computed from the binding constraints (ICP_{ih}) and (IRP_{il}).

The characterization of the equilibrium policies displays a combination of the screening and signaling effects. **Proposition 4** shows that the signaling effect reinforces the screening effect by reducing the policy for low-type lobbies; p_{hl}^* decreases due to the strategic complementarity and p_{hh}^* remains the same.

The pattern of tariff protection is quite different when lobbies have private information. It is easy to see that the policy distortions are asymmetric across different lobbies' types. While low types face signaling and screening costs, high types face only the latter cost. Information asymmetry makes "large" sectors relatively more protected than "small" sectors given their willingness to lobby. Moreover, from the screening effect, the policy maker obtains higher informational rents when she faces a high-type rival and lower rents when she faces a low-type rival. The amount of money given as contribution is more sensitive to the level of tariff protection (the slope of the contribution schedule is steeper), because the additional contribution given in response to an increase in protection from p_{il} to p_{ih} is more than the compensation for the welfare loss. Finally, since the expected protection falls, we have:

Corollary 2. The welfare of the equilibrium with informed lobbies is higher than the welfare of the equilibrium under perfect information.

The presence of private information within the lobby groups generates two information problems that reduce the lobbies' influence on the policy maker. Tariff protection decreases, imports increase and the welfare of the society increases.

5. Equilibrium discussion

We now discuss our results. We first identify the features of the model that interact with the information asymmetry to generate the information transmission problems. We then discuss the difference between our results and the ones found in the related literature. Finally, we discuss the SC and the consequent selected equilibrium.

As indicated earlier, we modified the GH model by introducing private information on the lobbies' preferences and substitutability of the goods (**Assumption 1**). Substitutability is a key feature of the model. Indeed, if $d = 0$ and $\gamma = 0$, the equilibrium policies would be the same as in the truthful equilibrium (since $\Psi = 0$ in **Proposition 2**) and the information asymmetry would not generate distortions in the political game. To be precise, when $d = 0$, the policy maker's preferences

are separable and the policy of one lobby does not affect the marginal deadweight loss from the policy (1) for the rival lobby. Lobbies' private information only distorts policies in the political game through the screening effect when the policy maker's preferences are not separable.

Non-separability of the policy maker's preferences is not a feature of our model only. Consider a model where the policy maker has to decide to which firm to allocate scarce resources (as in Esteban and Ray, 2006). Since resources are scarce, one unit allocated to one sector reduces the amount of resources available to other sectors, making the policy maker's preference non-separable. We can also think of many other examples where the policy maker's preference is not separable such as public goods or environmental laws. In all such examples, the screening effect will be naturally present if lobbies have private information.

The signaling effect comes from the private information nature in our model. Lobbies are privately informed and the policy maker is not. Since the policy maker cares about lobbies' profits (i.e., common value model), private information affects both lobbies' and policy maker's preferences.

For instance, if lobbies' private information is related to their internal organization cost and not to sectors' profits, there is no signaling effect. The signaling effect depends on lobbies' types directly affecting the marginal deadweight loss from the policy. In our model, its presence requires $\gamma > 0$ so that the second derivative of the profit function with respect to the policy depends on type. When $\gamma = 0$, the marginal deadweight loss from the policy (1) does not depend on the type because $\partial^2 \pi''(p) = 0$.

5.1. Rent distribution and the qualitative properties

One important result of our paper is the comparison of the total surplus distribution between the equilibrium of the political game with informed lobbies and the GH's truthful equilibrium. In the GH model, the policy maker does not extract any political rent when lobbies represent a negligible fraction of society (i.e., when the ownership of firms is highly concentrated). In our model, when a lobby faces a high-type rival, she leaves rents to the policy maker. Therefore, when lobbies are concentrated and have private information, the overall political rents are reduced while lobbies have to leave rents to the policy maker. The distribution of surplus in our model is therefore sharply different from that of the GH model.

Another interesting result of our model is the fact that information asymmetry is welfare enhancing, as shown in Corollaries 1 and 2. In contrast, Esteban and Ray (2006) show that the lobbies' private information reduces the society's welfare. Our results are different from theirs due to the policy maker's preferences and the role private information plays in our political game. They model the policy maker as a welfare minded agent who does not have the necessary information to make the correct decision (only the lobbies have this information). Lobbying is potentially positive since it may reveal useful information to the policy maker. However, the policy maker cannot make a good policy choice because a signal jamming problem prevents him from learning the lobbies' information. We model the policy maker as an agent who is willing to trade welfare for contributions. He knows the welfare maximizing policy (free-trade) and lobbying is essentially a rent-seeking activity that pushes policies away from the welfare optimum. The information the policy maker learns from lobbies informs him of the deadweight loss from shifting policies away from free-trade. This comparison stresses the nature of the results in Corollaries 1 and 2. We model lobbying as a rent-seeking activity, which is by construction a harmful activity for the society. Lobbies' private information impacts on the size of deadweight loss from a policy, but asymmetric information hinders an activity that harms society's welfare.

5.1.1. Selection criterion

The selection criterion (SC) restricts the possible conjectures each lobby has about the contribution schedule of her rival. This criterion helps us to select the most informative equilibrium. By "informative" we mean the equilibrium where all players endogenously learn the information of the others. Therefore, the equilibrium that satisfies the SC can be understood as an upper bound for the surplus that can be obtained in a game where players non-cooperatively learn each other's information, i.e., a decentralized equilibrium where the information is fully transmitted.

What about other types of equilibria that can emerge in this political game? The first one we can think of is the pooling equilibrium where different types of lobbies offer the same contribution and ask for the same policy. In such a case, the policy maker would not learn the lobbies' types and would not be able to screen any information from them. In a pooling equilibrium, different types of lobbies get the same policy. This policy cannot be optimal for both types though since each has a different willingness to pay for the policy. At least one type would gain from separating herself by sending a signal revealing her type to the policy maker and asking for a different policy. This intuitive argument suggests that a pooling equilibrium would not survive the intuitive criterion. In the online Appendix B, we characterize the contribution schedules that are interim efficient in the informed lobby problem. In particular, no pooling equilibria would survive the intuitive criterion.

Another type of equilibrium that could arise is one where lobbies offer a "pile" of contribution schedules that are conditional on a signal they would send to the policy maker after the contribution is accepted. Delaying information revelation could help the lobby relax the policy maker's incentive constraints, as shown by Maskin and Tirole (1990). In the online Appendix B, we show that such contribution schedules would give the lobbies the same payoffs as the contribution schedules that reveal the lobby's information directly, such as those considered in Section 3. The possibility of delaying information revelation does not give the lobbies any advantage compared to our selected equilibrium.

In summary, the SC helps us to characterize equilibrium contribution schedules that survive the intuitive criterion and are robust to delaying information revelation. They also define an upper bound for the political surplus obtained in a fully separating equilibrium.

6. Conclusion

We modified the GH model by assuming that the technology of productive sectors is the lobbies' private information. This new feature introduces private information on the lobbies' preferences into the political game and allows us to analyze the information transmission problem in the political game.

There are two information transmission problems in the political game. The first one is the signaling effect. The policy maker does not know the lobbies' types and the cost of tariff protection depends on these private informations. The lobbies of sectors that have higher welfare costs (more competitive sectors) may pretend they are less competitive in order to give a smaller contribution to the policy maker. The lobby of less competitive sectors then asks for less tariff protection in order to allow the policy maker to learn her true type through the schedule contributions he receives.

The second information transmission problem is the screening effect. The lobby of one sector does not know how much tariff protection her rival lobby is receiving. Since goods are substitutes (e.g., [Assumption 1](#)), lobbies do not know their marginal deadweight loss from tariff protection. In a separating equilibrium of the political game, lobbies understand that the policy maker learns the rival's type and design contribution schedules to screen this information out. Screening makes lobbies leave informational rents to the policy maker and ask for less tariff protection.

Both information transmission problems reduce lobbies' ability to influence the policy maker. Lobbies demand less tariff protection when compared to the perfect information equilibrium. We also derive results concerning the interaction between the signaling and screening effects within the bilateral relationship between a lobby and the policy maker. First, the signaling effect reduces the surplus of the political game and consequently mitigates the screening effect. Second, distortions from the screening effect make high-type lobbies reduce their contributions, leading to greater signaling distortions. The combination of the two effects generate greater distortions than each effect separately. Information transmission reduces lobbying activity and, therefore, increases society's welfare since lobbying is a rent-seeking activity (as in GH). Moreover, information transmission allows the policy maker to extract informational rents and also affects the surplus division between lobbies and policy maker in the political game.

The results of this paper raise some questions. The first is about transparency. It is commonly argued that transparency in the relationship between governments and lobbies is good for the society, which sharply contrasts with the results found here. Although the arguments in favor of transparency are traditionally based on the accountability of politicians (see [Coate and Morris, 1995](#)), not modeled in our paper, we showed that the absence of information asymmetry can harm the society. Therefore, our results suggest a trade-off between accountability and information transmission in political games.

The second question concerns the role of information transmission when policy makers use different policy instruments, such as non-tariff barriers. With non-tariff instruments, the government does not have tariff revenue and the deadweight loss of tariff protection is a function of the market elasticities and the sectors' import penetration, as pointed out by [Maggi and Rodrigues-Clare \(2000\)](#). A rich set of possibilities can arise in this case. The information transmission problems of our paper may be amplified or there may exist no information transmission at all. Therefore, different instruments should generate different effects given the information transmission problem.

Finally, the third question is how the size of the lobbying activity affects information transmission. A natural extension of our model is then to increase the number of lobbies in order to understand whether the information transmission problems increase or diminish with the lobbies' size in the political game. Such an exercise would be very complicated since the number of policies and states increase with the number of lobbies, making the model intractable. We conjecture that as the number of lobbies increases, the private information of one lobby becomes less relevant to the marginal cost of tariff protection of the other lobbies, rendering the screening effect less relevant. However, this limit argument does not suggest that these effects should be monotonic. It is perfectly conceivable that the complexity of the information transmission problem may increase substantially with a small number of lobbies.

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Appendix A

Proof of Lemma 1. If the contribution schedules satisfy the SC, then they will satisfy binding constraints (ICP_{ih}) and (IRP_{il}). Constraint (ICP_{ih}) is

$$C_{ih} + C_{hi} + \lambda W(\theta_i, p_{ih}, \theta_h, p_{hi}) \geq C_{il} + C_{hi} + \lambda W(\theta_i, p_{il}, \theta_h, p_{hi})$$

or

$$C_{ih} - C_{il} \geq \lambda [W(\theta_i, p_{il}, \theta_h, \mathbf{p}_{hi}) - W(\theta_i, p_{ih}, \theta_h, \mathbf{p}_{hi})]. \quad (\text{A.1})$$

Constraint (ICP_{il}) is

$$C_{il} + \mathbf{C}_{li} + \lambda W(\theta_i, p_{il}, \theta_l, \mathbf{p}_{li}) \geq C_{ih} + \mathbf{C}_{li} + \lambda W(\theta_i, p_{ih}, \theta_l, \mathbf{p}_{li})$$

or

$$C_{ih} - C_{il} \leq \lambda [W(\theta_i, p_{il}, \theta_l, \mathbf{p}_{li}) - W(\theta_i, p_{ih}, \theta_l, \mathbf{p}_{li})]. \quad (\text{A.2})$$

Putting (A.1) and (A.2) together, we get

$$W(\theta_i, p_{il}, \theta_l, \mathbf{p}_{li}) - W(\theta_i, p_{ih}, \theta_l, \mathbf{p}_{li}) \geq W(\theta_i, p_{il}, \theta_h, \mathbf{p}_{hi}) - W(\theta_i, p_{ih}, \theta_h, \mathbf{p}_{hi}).$$

This last inequality can be written as

$$\int_{p_{ih}}^{p_{il}} \frac{\partial W}{\partial p^1}(\theta_i, s, \theta_l, \mathbf{p}_{li}) ds - \int_{p_{ih}}^{p_{il}} \frac{\partial W}{\partial p^1}(\theta_i, s, \theta_h, \mathbf{p}_{hi}) ds \geq 0$$

or

$$\int_{p_{ih}}^{p_{il}} \left[\int_{\mathbf{p}_{hi}}^{\mathbf{p}_{li}} \frac{\partial^2 W}{\partial p^1 \partial p^2}(\theta_i, s, \theta_l, \tilde{\mathbf{p}}) d\tilde{\mathbf{p}} + \int_{\theta_h}^{\theta_l} \frac{\partial^2 W}{\partial \theta^2 \partial p^1}(\theta_i, s, \tilde{\theta}, \mathbf{p}_{hi}) d\tilde{\theta} \right] ds \geq 0.$$

From (1) we have that

$$\begin{aligned} \frac{\partial^2 W}{\partial \theta^2 \partial p^1}(\theta^1, p^1, \theta^2, \mathbf{p}^2) &= 0 \quad \text{and} \\ \frac{\partial^2 W}{\partial p^1 \partial p^2}(\theta^1, p^1, \theta^2, \mathbf{p}^2) &= d. \end{aligned}$$

Since $\mathbf{p}_{hi} \geq \mathbf{p}_{li}$, $\int_{p_{ih}}^{p_{il}} \left[\int_{\mathbf{p}_{hi}}^{\mathbf{p}_{li}} \frac{\partial^2 W}{\partial p^1 \partial p^2}(\theta_i, s, \theta_l, \tilde{\mathbf{p}}) d\tilde{\mathbf{p}} \right] ds \geq 0$ if and only if $p_{ih} \geq p_{il}$, which proves the lemma. $\square$

Proof of Lemma 2. Provided that constraints (IRP_{il}) and (ICP_{ih}) are binding, contributions are defined as

$$C_{il} = -\mathbf{C}_{li} - \lambda [W(\theta_i, p_{il}, \theta_l, \mathbf{p}_{li}) - W(\theta_i, p^e, \theta_l, p^e)] \quad (\text{A.3})$$

and

$$C_{ih} = C_{il} - \lambda [W(\theta_i, p_{ih}, \theta_h, \mathbf{p}_{hi}) - W(\theta_i, p_{il}, \theta_h, \mathbf{p}_{hi})]. \quad (\text{A.4})$$

We plug these contributions into the lobby's utility function in (6) to get

$$\begin{aligned} & \max_{p_{ih}, p_{il}} z \{ \theta_i \pi(p_{ih}) + \lambda [W(\theta_i, p_{ih}, \theta_h, \mathbf{p}_{hi}) - W(\theta_i, p^e, \theta_l, p^e)] \} \\ & + (1-z) \{ \theta_i \pi(p_{il}) + \lambda [W(\theta_i, p_{il}, \theta_l, \mathbf{p}_{li}) - W(\theta_i, p^e, \theta_l, p^e)] \} \\ & - z\lambda [W(\theta_i, p_{il}, \theta_h, \mathbf{p}_{hi}) + W(\theta_i, p_{il}, \theta_l, \mathbf{p}_{li})] + \mathbf{C}_{li} \\ & \text{s.t. (IC}_{-i}). \end{aligned}$$

The first-order conditions of this problem are

$$\theta_i \pi'(p_{ih}) - \lambda \frac{\partial W}{\partial p^1}(\theta_i, p_{ih}, \theta_h, \mathbf{p}_{hi}) - \mu_{-i} \left[\theta_{-i} \pi'(p_{ih}) - \lambda \frac{\partial W}{\partial p^1}(\theta_i, p_{ih}, \theta_h, \mathbf{p}_{hi}) \right] = 0, \quad (\text{A.5})$$

and

$$\begin{aligned} & \theta_i \pi'(p_{il}) - \lambda \frac{\partial W}{\partial p^1}(\theta_i, p_{il}, \theta_l, \mathbf{p}_{li}) + \frac{z\lambda d}{1-z} [(\mathbf{p}_{li} - p^e) - (\mathbf{p}_{lh} - p^e)] \\ & - \mu_{-i} \left[\theta_{-i} \pi'(p_{il}) - \lambda \frac{\partial W}{\partial p^1}(\theta_i, p_{il}, \theta_l, \mathbf{p}_{li}) \right] - \mu_{-i} \frac{z\lambda d}{1-z} [(\mathbf{p}_{li} - p^e) - (\mathbf{p}_{lh} - p^e)] = 0. \end{aligned} \quad (\text{A.6})$$

We add and subtract the terms $\mu_{-i}\theta_i\pi'(p_{ih})$ and $\mu_{-i}\theta_i\pi'(p_{il})$, respectively, in (A.5) and (A.6), and divide both by $(1 - \mu_{-i})$ to get

$$\theta_i\pi'(p_{ih}) - \lambda \frac{\partial W}{\partial p^1}(\theta_i, p_{ih}, \theta_h, \mathbf{p}_{hi}) + \frac{\mu_{-i}}{1 - \mu_{-i}} \Delta\theta_i\pi'(p_{ih}) = 0,$$

and

$$\theta_i\pi'(p_{il}) - \lambda \frac{\partial W}{\partial p^1}(\theta_i, p_{il}, \theta_l, \mathbf{p}_{li}) + \frac{\mu_{-i}}{1 - \mu_{-i}} \Delta\theta_i\pi'(p_{il}) + \frac{z\lambda d}{1 - z}(\mathbf{p}_{li} - \mathbf{p}_{lh}) = 0,$$

which imply condition (8).

Moreover, we have the usual Kuhn–Tucker conditions,

$$\begin{aligned} \mu_{-i} &\geq 0 \quad \text{for all } -i, \\ E[\theta_{-i}\pi(p_{-i}) - C_{-i}] - E[\theta_{-i}\pi(p_i) - C_i] &\geq 0 \quad \text{for all } i, -i, \text{ and} \\ (E[\theta_{-i}\pi(p_i) - C_i] - E[\theta_{-i}\pi(p_{-i}) - C_{-i}])\mu_{-i} &= 0 \quad \text{for all } i, -i. \quad \square \end{aligned}$$

Proof of Lemma 3. Let $p_{ik} = p(\theta_i, \theta_k)$ and contributions C_{ik} , computed from (A.3) and (A.4) for every k , be a solution of problem (6) satisfying the SC.

We will identify cases where the constraint (IC_{-i}) is not binding. For the type- $(-i)$ informed lobby problem, this constraint can be written as

$$E[V(\theta_i, p(\theta_i, \cdot), C(\theta_i, p(\theta_i, \cdot))) - E[V(\theta_i, p(\theta_{-i}, \cdot), C(\theta_{-i}, p(\theta_{-i}, \cdot)))] \geq 0.$$

We will artificially write this inequality as if lobbies could lie to the policy maker and announce any type $\tilde{\theta} \in [\theta_l, \theta_h]$. That is, we will extend types to be continuous variables, which will simplify the proof.

The policies for the intermediate values of $\tilde{\theta}$, $p(\tilde{\theta}, \cdot)$, are defined by applying the Implicit Function Theorem on (8), where we replace θ_i by $\tilde{\theta}$, and μ_{-i} by 0. That is, $p(\tilde{\theta}, \cdot)$ must satisfy

$$\tilde{\theta}\pi'(p(\tilde{\theta}, \theta_k)) - \lambda \frac{\partial W}{\partial p^1}(\tilde{\theta}, p(\tilde{\theta}, \theta_k), \theta_k, \mathbf{p}_{ki}) + \lambda I(k) \frac{z}{1 - z} d(\mathbf{p}_{li} - \mathbf{p}_{hi}) = 0.$$

Notice that we can apply the Implicit Function Theorem since, from Assumption 2, program (6) is strictly concave.

The contribution when the rival lobby is the low type, C_{il} , will be denoted by $C_l(\tilde{\theta}, p(\tilde{\theta}, \theta_l))$, and is computed from (A.3) by replacing p_{il} with $p(\tilde{\theta}, \theta_l)$. The contribution when the rival is the high type, C_{ih} , is denoted by $C_h(\tilde{\theta}, p(\tilde{\theta}, \theta_h), p(\tilde{\theta}, \theta_l))$ and is computed from (A.4) by replacing p_{ik} with $p(\tilde{\theta}, \theta_k)$, for every k . It follows that $p(\tilde{\theta}, \cdot)$ is a differentiable function of $\tilde{\theta}$ and the contributions are differentiable functions of $\tilde{\theta}$ and $p(\cdot, \theta_k)$, for every k .

The constraint (IC_{-i}) can be written equivalently as

$$E \left[\int_{\theta_{-i}}^{\theta_i} \frac{\partial V}{\partial \tilde{\theta}}(\theta_i, p(\tilde{\theta}, \cdot), C_l(\tilde{\theta}, p(\tilde{\theta}, \cdot), p(\tilde{\theta}, \cdot))) d\tilde{\theta} \right] \geq 0, \quad (\text{A.7})$$

where the expectation is taken with respect to the rival's type. Notice that the announced type is different from the lobby's true type.

In turn, we have that

$$\begin{aligned} \frac{dV}{d\tilde{\theta}}(\theta_i, p(\tilde{\theta}, \theta_h), C_h) &= \left(\theta_i\pi'(p(\tilde{\theta}, \theta_h)) - \frac{\partial C_h}{\partial p^1} \right) \frac{\partial p}{\partial \tilde{\theta}}(\tilde{\theta}, \theta_h) - \frac{\partial C_h}{\partial \tilde{\theta}} - \frac{\partial C_h}{\partial p^2} \frac{\partial p}{\partial \tilde{\theta}}(\tilde{\theta}, \theta_l) \quad \text{and} \\ \frac{dV}{d\tilde{\theta}}(\theta_i, p(\tilde{\theta}, \theta_l), C_l) &= \left(\theta_i\pi'(p(\tilde{\theta}, \theta_l)) - \frac{\partial C_l}{\partial p^1} \right) \frac{\partial p}{\partial \tilde{\theta}}(\tilde{\theta}, \theta_l) - \frac{\partial C_l}{\partial \tilde{\theta}}, \end{aligned}$$

where we suppressed the arguments of the contribution function for simplicity.

Since $p(\tilde{\theta}, \cdot)$ satisfies condition (8) for every $\tilde{\theta}$, we have that

$$\tilde{\theta}\pi'(p(\tilde{\theta}, \theta_h)) = \frac{\partial C_h}{\partial p^1}(\tilde{\theta}, p(\tilde{\theta}, \theta_h), p(\tilde{\theta}, \theta_l)) = -\lambda \frac{\partial W}{\partial p^1}(\tilde{\theta}, p(\tilde{\theta}, \theta_h), \theta_h, \mathbf{p}_{hi}) \quad \text{and}$$

$$\begin{aligned} \tilde{\theta}\pi'(p(\tilde{\theta}, \theta_l)) &= \frac{\partial C_l}{\partial p^1}(\tilde{\theta}, p(\tilde{\theta}, \theta_l)) + \frac{z}{1 - z} \frac{\partial C_h}{\partial p^2}(\tilde{\theta}, p(\tilde{\theta}, \theta_h), p(\tilde{\theta}, \theta_l)) \\ &= -\lambda \frac{\partial W}{\partial p^1}(\tilde{\theta}, p(\tilde{\theta}, \theta_l), \theta_l, \mathbf{p}_{li}) - \lambda \frac{z}{1 - z} \left(\frac{\partial W}{\partial p^1}(\tilde{\theta}, p(\tilde{\theta}, \theta_l), \theta_l, \mathbf{p}_{li}) - \frac{\partial W}{\partial p^1}(\tilde{\theta}, p(\tilde{\theta}, \theta_l), \theta_h, \mathbf{p}_{hi}) \right). \end{aligned}$$

Hence, the derivatives of the lobby's utility with respect to the policy simplify to

$$\begin{aligned}\frac{dV}{d\tilde{\theta}}(\theta_i, p(\tilde{\theta}, \theta_h), C_h) &= (\theta_i - \tilde{\theta})\pi'(p(\tilde{\theta}, \theta_h))\frac{\partial p}{\partial \tilde{\theta}} - \frac{\partial C_h}{\partial \tilde{\theta}} - \frac{\partial C_h}{\partial p^1}\frac{\partial p}{\partial \tilde{\theta}} \quad \text{and} \\ \frac{dV}{d\tilde{\theta}}(\theta_i, p(\tilde{\theta}, \theta_l), C_l) &= (\theta_i - \tilde{\theta})\pi'(p(\tilde{\theta}, \theta_l))\frac{\partial p}{\partial \tilde{\theta}} - \frac{\partial C_l}{\partial \tilde{\theta}} + \frac{z}{1-z}\frac{\partial C_h}{\partial p}\frac{\partial p}{\partial \tilde{\theta}}.\end{aligned}$$

Substituting them back into condition (A.7) gives

$$E\left[\int_{\theta_{-i}}^{\theta_i} (\theta_i - \tilde{\theta})\pi'(p(\tilde{\theta}, \cdot))\frac{\partial p}{\partial \tilde{\theta}}(\tilde{\theta}, \cdot) - \frac{\partial C}{\partial \tilde{\theta}}d\tilde{\theta}\right] \geq 0. \quad (\text{A.8})$$

To find the expression for the direct derivative of the contribution with respect to $\tilde{\theta}$, $\frac{\partial C}{\partial \tilde{\theta}}$, we have to look at the welfare functions given by

$$\begin{aligned}W(\tilde{\theta}, p(\tilde{\theta}, \theta_k), \theta_k, \mathbf{p}_{ki}) &= A - \frac{b}{2}(p(\tilde{\theta}, \theta_k) - p^e)^2 + d(p(\tilde{\theta}, \theta_k) - p^e)(\mathbf{p}_{ki} - p^e) \\ &\quad - \frac{b}{2}(\mathbf{p}_{ki} - p^e)^2 + \tilde{\theta}\pi(p(\tilde{\theta}, \theta_k)) - \tilde{\theta}\pi'(p(\tilde{\theta}, \theta_k))(p(\tilde{\theta}, \theta_k) - p^e) \\ &\quad + \theta_k\pi(\mathbf{p}_{ki}) - \theta_k\pi'(\mathbf{p}_{ki})(\mathbf{p}_{ki} - p^e),\end{aligned}$$

where A is a constant that depends only on the parameters a, b and d . The derivatives of the welfare function are then given by

$$\frac{\partial W}{\partial \tilde{\theta}}(\tilde{\theta}, p(\tilde{\theta}, \theta_k), \theta_k, \mathbf{p}_{ki}) = \pi(p(\tilde{\theta}, \theta_k)) - \pi'(p(\tilde{\theta}, \theta_k))(p(\tilde{\theta}, \theta_k) - p^e), \quad (\text{A.9})$$

which implies that

$$\frac{\partial W}{\partial \tilde{\theta}}(\tilde{\theta}, p, \theta_k, \mathbf{p}) = \pi(p^e). \quad (\text{A.10})$$

(i) For $\gamma = 0$, the profit function is $\pi(p) = p$. Since contributions are computed from (A.3) and (A.4), Eq. (A.10) implies that $\partial C_l/\partial \tilde{\theta} = \partial C_h/\partial \tilde{\theta} = 0$. Therefore, condition (A.8) becomes

$$E\left[\int_{\theta_{-i}}^{\theta_i} (\theta_i - \tilde{\theta})\frac{\partial p}{\partial \tilde{\theta}}(\tilde{\theta}, \cdot)d\tilde{\theta}\right] \geq 0.$$

If $\theta_i = \theta_h$, this last inequality holds when

$$E\left[(\theta_h - \tilde{\theta})\frac{\partial p}{\partial \tilde{\theta}}(\tilde{\theta}, \cdot)\right] \geq 0,$$

for all $\tilde{\theta}$, which is true since $\frac{\partial p}{\partial \tilde{\theta}}(\tilde{\theta}, \cdot) \geq 0$. Likewise, it is easy to check this inequality also holds for $\theta_i = \theta_l$. Therefore, the lobby is always better off telling the truth and constraints (IC_{-i}) , for $i = h, l$, are not binding when $\gamma = 0$. This proves (i).

(ii) For $\gamma = 1$, the profit function is $\pi(p) = \frac{p^2}{2}$. Contributions are also computed from (A.3) and (A.4). From (A.9) and (A.10) we have that

$$\frac{\partial C_l}{\partial \tilde{\theta}} = \frac{1}{2}(p(\tilde{\theta}, \theta_k) - p^e)^2.$$

Hence, when $-i = h$, condition (A.8) can be written as

$$E\left[\int_{\theta_h}^{\theta_l} \left[(\theta_l - \tilde{\theta})p(\tilde{\theta}, \cdot)\frac{\partial p}{\partial \tilde{\theta}}(\tilde{\theta}, \cdot) - \frac{1}{2}(p(\tilde{\theta}, \cdot) - p^e)^2\right]d\tilde{\theta}\right] \geq 0$$

or

$$E\left[\int_{\theta_l}^{\theta_h} \left[(\tilde{\theta} - \theta_l)p(\tilde{\theta}, \cdot)\frac{\partial p}{\partial \tilde{\theta}}(\tilde{\theta}, \cdot) + \frac{1}{2}(p(\tilde{\theta}, \cdot) - p^e)^2\right]d\tilde{\theta}\right] \geq 0.$$

By the Implicit Function Theorem we have that $\frac{\partial p}{\partial \tilde{\theta}}(\tilde{\theta}, \cdot) \geq 0$. Since $\tilde{\theta} \geq \theta_l$, the above inequality holds. This implies that constraint (IC_l) is not binding. This proves (ii).

Notice that for the high-type lobby we have that

$$E \left[\int_{\theta_l}^{\theta_h} (\theta_h - \tilde{\theta}) p(\tilde{\theta}, \cdot) \frac{\partial p}{\partial \tilde{\theta}}(\tilde{\theta}, \cdot) - \frac{1}{2} (p(\tilde{\theta}, \cdot) - p^e)^2 \right] d\tilde{\theta},$$

which clearly may not be positive since $p(\theta_h, \cdot) - p^e > 0$ and $\frac{\partial p}{\partial \tilde{\theta}}(\tilde{\theta}, \cdot) \geq 0$. $\square$

Proof of Proposition 1. First let us prove the following useful lemma. Let $F, G : \mathbb{R}_+^{n+1} \rightarrow \mathbb{R}$ be twice differentiable and additively separable in the n first arguments such that

$$F_j, G_j > 0, G_{n+1} < 0, \partial_j \left(\frac{F_j}{G_j} \right) < 0 \quad \text{and} \quad \partial_{n+1} \left(\frac{F_j}{G_j} \right) \geq 0,$$

for all $j = 1, \dots, n$, where the sub index j denotes the partial derivative with respect to the j -th variable. Consider the problem:

$$\begin{aligned} \max_{v \in D} F(v, \delta) \\ \text{s.t. } G(v, \delta) = 0, \end{aligned} \tag{A.11}$$

where D is a convex subset of $\mathbb{R}_+^{n+1}$. Suppose that problem (A.11) has a solution.

Lemma 4. The solution of problem (A.11), $v^*(\delta)$, is strictly increasing in δ and the Lagrangian multiplier of its constraint, μ , is strictly decreasing in δ .

Proof. The hypotheses of the lemma imply that the Jacobean of G satisfies $|J_v G| = \prod G_j > 0$ (in particular, at the optimum solution $v^*(\delta)$ for parameter δ). By the Lagrange Theorem, there exists a Lagrangian multiplier μ for problem (A.11).

The first-order condition of the Lagrangian gives

$$\mu = \frac{F_j}{G_j},$$

where F_j and G_j are computed at $(v^*(\delta), \delta)$. By the Implicit Function Theorem, $v^*(\delta)$ is a differentiable function of δ . Taking the implicit derivative, we get

$$\frac{d\mu}{d\delta} = \frac{1}{G_j} \left[(F_{jj} - \mu G_{jj}) \frac{dv_j^*}{d\delta} + F_{j(n+1)} - \mu G_{j(n+1)} \right],$$

where we are omitting the dependence of the functions on their arguments. Notice that

$$\begin{aligned} F_{jj} - \mu G_{jj} < 0 \quad \text{if and only if} \quad \partial_j \left(\frac{F_j}{G_j} \right) < 0 \quad \text{and} \\ F_{j(n+1)} - \mu G_{j(n+1)} \geq 0 \quad \text{if and only if} \quad \partial_{n+1} \left(\frac{F_j}{G_j} \right) \geq 0, \end{aligned}$$

which is true by hypothesis. Since $G(v^*(\delta), \delta) = 0$, by taking the implicit derivative we have that $\sum_i G_j \frac{dv_j^*}{d\delta} = -G_{n+1} > 0$. By the hypothesis on the sign of the derivatives of G , $\frac{dv_j^*}{d\delta} > 0$, for some j . Therefore, $\frac{d\mu}{d\delta} < 0$ and, using the above expression of $\frac{d\mu}{d\delta}$, we conclude that $\frac{dv_j^*}{d\delta} > 0$, for all j , as we want to prove. $\square$

Let us come back to the proof of Proposition 1.

(i) Let $D = (-\infty, \bar{p}_l] \times (-\infty, \bar{p}_h] \times \mathbb{R}$, $v = (p_l, p_h)$, $\delta = \bar{U}$, $F(v, \delta) = E[V(\theta_l, p_l, C_l(p_l, \cdot))]$ and $G(v, \delta) = E[V(\theta_h, p_l, C_l(p_l, \cdot))] - \bar{U}$. By the definition of V , notice that F and G are additively separable in (p_l, p_h) . Moreover,

$$\frac{1}{1-z} F_{p_l} = \frac{1}{1-z} G_{p_l} - \Delta \theta_h p_l = \theta_l p_l - \lambda(b + \theta_l)(p_l - p^e) + \lambda d(\mathbf{p}_l - p^e) - \lambda d \frac{z}{1-z} [\mathbf{p}_h - \mathbf{p}_l],$$

and

$$\frac{1}{z} F_{p_h} = \frac{1}{z} G_{p_h} - \Delta \theta_h p_h = \theta_l p_h - \lambda(b + \theta_l)(p_h - p^e) + \lambda d(\mathbf{p}_h - p^e).$$

Hence, $F_{p_{li}}, F_{p_{lh}}, G_{p_{li}}, G_{p_{lh}} > 0, G_{\bar{U}} < 0$,

$$\frac{1}{1-z} \partial_{p_{ll}} \left(\frac{F_{p_{ll}}}{G_{p_{ll}}} \right) = \frac{(\theta_l - \lambda(b + \theta_l))G_{p_{ll}} - (\theta_h - \lambda(b + \theta_l))F_{p_{ll}}}{(G_{p_{ll}})^2} \leq -\frac{\Delta\theta_h}{G_{p_{ll}}} < 0,$$

and

$$\partial_{\bar{U}} \left(\frac{F_{p_{ll}}}{G_{p_{ll}}} \right) = 0.$$

Analogously, we have that

$$\partial_{p_{lh}} \left(\frac{F_{p_{lh}}}{G_{p_{lh}}} \right) < 0 \quad \text{and} \quad \partial_{\bar{U}} \left(\frac{F_{p_{lh}}}{G_{p_{lh}}} \right) = 0.$$

Applying Lemma 4, we have that the optimal policies p_{lh} and p_{ll} are strictly increasing in $\bar{U}$ and the multiplier μ is strictly decreasing in $\bar{U}$.

Policies p_{lh} and p_{ll} are increasing because of the direct effect of signaling. To relate this result to the screening effect, we must look at the best-response policies:

$$p_{lh}(\mathbf{p}_{hl}) = \frac{(\theta_l - \frac{\mu\Delta\theta_h}{1-\mu})p^e + \lambda d(\mathbf{p}_{hl} - p^e)}{\lambda(b + \theta_l) - \theta_l + \frac{\mu\Delta\theta_h}{1-\mu}} + p^e \quad \text{and}$$

$$p_{ll}(\mathbf{p}_{hl}, \mathbf{p}_{ll}) = \frac{(\theta_l - \frac{\mu\Delta\theta_h}{1-\mu})p^e + \lambda d(\mathbf{p}_{ll} - p^e) - \frac{z}{1-z} \lambda d(\mathbf{p}_{hl} - \mathbf{p}_{ll})}{\lambda(b + \theta_l) - \theta_l + \frac{\mu\Delta\theta_h}{1-\mu}} + p^e.$$

Notice that the distortion term due to screening that appears in the best-response function p_{ll} is given by

$$\frac{z}{1-z} \frac{\lambda d(\mathbf{p}_{hl} - \mathbf{p}_{ll})}{\lambda(b + \theta_l) - \theta_l + \frac{\mu\Delta\theta_h}{1-\mu}},$$

which is clearly decreasing with respect to μ and, consequently, increasing with respect to $\bar{U}$.

In turn, let us compute the effects of $\bar{U}$ on the information rent. The information rent is the difference between the contribution C_{lh} and the contribution a lobby would give if she knew the rival lobby's private information. This last contribution can be computed from the individual rationality constraint (IRP_{lh}). We explicitly have the informational rent as

$$\lambda[W(\theta_i, p_{il}, \theta_h, \mathbf{p}_{hi}) - W(\theta_i, p_{il}, \theta_l, \mathbf{p}_{li})] - \lambda[W(\theta_i, p^e, \theta_h, p^e) - W(\theta_i, p^e, \theta_l, p^e)].$$

Given the functional form of the welfare function, this last expression becomes

$$-\frac{b + \theta_h}{2}(\mathbf{p}_{hl} - p^e)^2 + \frac{b + \theta_l}{2}(\mathbf{p}_{ll} - p^e)^2 + d(p_{ll} - p^e)(\mathbf{p}_{hl} - \mathbf{p}_{ll}),$$

which is clearly increasing in p_{ll} since $(\mathbf{p}_{hl} - \mathbf{p}_{ll})$ is positive from the SC condition. Therefore, the information rent is increasing with respect to $\bar{U}$, provided that p_{ll} is increasing with respect to $\bar{U}$.

(ii) Let $D = (-\infty, p^e]^2 \times [0, 1]$, $v = (p_{ll}, p_{lh})$, $\delta = 1 - z$, $F(v, \delta) = E^z[V(\theta_l, p_l, C_l(p_l, \cdot))]$ and $G(v, \delta) = E^z[V(\theta_h, p_l, C_l(p_l, \cdot)) - \bar{U}]$. Notice that F and G are additively separable in (p_{ll}, p_{lh}) . Like item (i), we have that $F_{p_{li}}, G_{p_{li}} > 0$, and

$$\partial_{p_{li}} \left(\frac{F_{p_{li}}}{G_{p_{li}}} \right) < 0 \quad \text{and} \quad \partial_{\delta} \left(\frac{F_{p_{li}}}{G_{p_{li}}} \right) = 0,$$

for $i = l, h$. Now

$$-G_{\delta} = V(\theta_h, p_{lh}, C_l(p_{lh}, \theta_h)) - V(\theta_h, p_{ll}, C_l(p_{ll}, \theta_l)) > 0.$$

Applying Lemma 4, we have that the optimal policies p_{ll} and p_{lh} are strictly decreasing in z and the multiplier μ is strictly increasing in z . Therefore, z is related to more screening distortion and also a greater shadow price of the signaling constraint (IC_h). $\square$

Proof of Theorem 1. First let us prove the following useful lemma. Given parameters $\alpha_j, \beta_j, r > 0$ and $\bar{v}_j \in \mathbb{R}$, $j = 1, 2$, consider the following maximization problem:

$$\begin{aligned} \max_{(v_1, v_2) \in \mathbb{R}^2} & -\frac{\alpha_1}{2}(v_1)^2 - \frac{\alpha_2}{2}(v_2)^2 \\ \text{s.t.} & \frac{\beta_1}{2}(v_1 - \bar{v}_1)^2 + \frac{\beta_2}{2}(v_2 - \bar{v}_2)^2 \geq r. \end{aligned} \tag{A.12}$$

Lemma 5. Suppose that $\bar{v}_1 \neq 0$ and $\bar{v}_2 \neq 0$. There exists a unique solution of problem (A.12) which is a continuous function of the parameters $(\alpha_1, \alpha_2, \beta_1, \beta_2, \bar{v}_1, \bar{v}_2, r)$.

Proof. Let us add the constraint $\frac{\beta_1}{2}(v_1 - \bar{v}_1)^2 + \frac{\beta_2}{2}(v_2 - \bar{v}_2)^2 \leq \tilde{r}$ to problem (A.12), for some $\tilde{r} > 0$ sufficiently large such that its unconstrained solution, $(v_1, v_2) = (0, 0)$, satisfies it. We can easily see that the solution of the original problem is the same as in the transformed problem. Moreover, the transformed problem now has a constraint set which is a continuous and compact-valued correspondence of the parameters, and the objective function is also a continuous function of the parameters. By the maximum theorem, the solution of problem (A.12) defines an upper semi-continuous correspondence of the parameters.

Let us prove that the solution is unique. It is easy to see that the constraint of problem (A.12) satisfies the usual regularity condition of the Kuhn–Tucker Theorem. Hence, there exists a Lagrangian multiplier $\mu \in \mathbb{R}$ such that the solution of problem (A.12) satisfies the necessary first- and second-order conditions of the Lagrangian:

$$L(v, \mu) = -\frac{\alpha_1}{2}(v_1)^2 - \frac{\alpha_2}{2}(v_2)^2 + \mu \left[\frac{\beta_1}{2}(v_1 - \bar{v}_1)^2 + \frac{\beta_2}{2}(v_2 - \bar{v}_2)^2 - r \right]$$

and the usual slackness condition $\mu \left[\frac{\beta_1}{2}(v_1 - \bar{v}_1)^2 + \frac{\beta_2}{2}(v_2 - \bar{v}_2)^2 - r \right] = 0$.

The first- and second-order conditions of the Lagrangian are

$$0 = \frac{\partial L}{\partial v_j}(v, \mu) = -\alpha_j v_j + \mu \beta_j (v_j - \bar{v}_j) \quad \text{and}$$

$$0 \geq \frac{\partial^2 L}{\partial v_j^2}(v, \mu) = \mu \beta_j - \alpha_j,$$

for all $j = 1, 2$. Let us consider the following cases:

1. $\mu = 0$. Then, $(v_1, v_2) = (0, 0)$ satisfies the first- and second-order conditions. This is indeed the solution when feasible for problem (A.12).
2. $\mu \neq 0$. Then, the constraint of problem (A.12) is binding. Since $\bar{v}_1 \neq 0$ and $\bar{v}_2 \neq 0$, then $v_j = \frac{\mu \beta_j \bar{v}_j}{\mu \beta_j - \alpha_j}$ and $\alpha_j > \mu \beta_j$, for $j = 1, 2$. Substituting the value of v_1 and v_2 into the slackness condition we get

$$\frac{\beta_1}{2} \left(\frac{\bar{v}_1 \alpha_1}{\mu \beta_1 - \alpha_1} \right)^2 + \frac{\beta_2}{2} \left(\frac{\bar{v}_2 \alpha_2}{\mu \beta_2 - \alpha_2} \right)^2 = r.$$

Defining the function $f(\bar{v}, s, \mu) = \frac{\bar{v}}{\mu s - 1}$ for $\mu < s^{-1}$, where $s > 0$, we can see that the previous condition can be written as

$$\beta_1 f(\bar{v}_1, \beta_1/\alpha_1, \mu) + \beta_2 f(\bar{v}_2, \beta_2/\alpha_2, \mu) = 2r.$$

Suppose, without loss of generality, that $\bar{v}_j > 0$, $j = 1, 2$. Then, $f(\bar{v}, s, \mu)$ is strictly decreasing in μ , for all $\mu < s^{-1}$. Therefore, there must exist at most one solution of the previous equation.

We then conclude that the solution is an upper semi-continuous single-valued correspondence (i.e., continuous function) of the parameters. $\square$

Now we are ready to prove the result. Let us consider the best-response problems of the informed lobby problem.

Substituting (ICP_{ik}) and (IRP_{ik}) for $k = l$ and $i = h$ in the objective function of high-type lobby, we can eliminate the contribution variables from the high-type informed lobby problem:

$$\max_{p_{hh}, p_{hl}} Q^h(p_{hh}, p_{hl}) + C_{lh},$$

where

$$Q^h(p_{hh}, p_{hl}) = \theta_h [z\pi(p_{hh}) + (1-z)\pi(p_{hl})] + \lambda z [W(\theta_h, p_{hh}, \theta_h, \mathbf{p}_{hh}) - W(\theta_h, p_{hl}, \theta_h, \mathbf{p}_{hh}) \\ + W(\theta_h, p_{hl}, \theta_l, \mathbf{p}_{lh}) - W(\theta_h, p^e, \theta_l, p^e)] + \lambda (1-z) [W(\theta_h, p_{hl}, \theta_l, \mathbf{p}_{lh}) - W(\theta_h, p^e, \theta_l, p^e)]$$

is a quadratic and concave function of (p_{hh}, p_{hl}) (by Assumption 2 and the hypothesis that $\gamma \in \{0, 1\}$). We can then apply Lemma 5 (for the case where the constraint is not binding) to show that the argument that maximizes this problem is a continuous function of the parameters including the policies of the rival lobby.

Rearranging (ICP_{ik}) and (IRP_{ik}) , we can eliminate the contributions from the informed lobby problem, which becomes

$$\begin{aligned} \max_{p_{lh}, p_{ll}} Q^l(p_{lh}, p_{ll}) + C_{ll} \\ \text{s.t. } P(p_{lh}, p_{ll}) \leq E[V(\theta_h, p_h, C_h)] \end{aligned}$$

where

$$\begin{aligned} Q^l(p_{lh}, p_{ll}) = \theta_l [z\pi(p_{lh}) + (1-z)\pi(p_{ll})] + \lambda z [W(\theta_l, p_{lh}, \theta_h, \mathbf{p}_{hl}) - W(\theta_l, p_{ll}, \theta_h, \mathbf{p}_{hl}) \\ + W(\theta_l, p_{ll}, \theta_l, \mathbf{p}_{ll}) - W(\theta_l, p^e, \theta_l, p^e)] + \lambda(1-z) [W(\theta_l, p_{ll}, \theta_l, \mathbf{p}_{ll}) - W(\theta_h, p^e, \theta_l, p^e)] \end{aligned}$$

and $P(p_{lh}, p_{ll}) = Q^l(p_{lh}, p_{ll}) + C_{ll} + z\Delta\theta\pi(p_{lh}) + (1-z)\Delta\theta\pi(p_{ll})$ are quadratic and concave functions of (p_{lh}, p_{ll}) (by [Assumption 2](#) and the hypothesis that $\gamma \in \{0, 1\}$). Notice that the unconstrained maximum of Q^l and P are different coordinate by coordinate, because $\pi'(\cdot) > 0$. We can then apply [Lemma 5](#) to show that the argument that maximizes this problem is a continuous function of the parameters including the policies of the rival lobby.

We have just argued that the lobbies' best-responses in the political game are continuous functions of the rival lobbies' policies. It only remains to define the set of policies where these best responses are onto functions. Take the compact square $K = [0, a/(b-d)]^2$ in $\mathbb{R}^2$ as the lobbies's strategy space. For the type- i lobby, if the set of strategies (p_{ih}, p_{il}) of her rival is K , her best response will also be in K from the definition of the demand curve in the text. Therefore, the best responses map K onto K and¹⁸ the existence of an equilibrium is an immediate consequence of the Brouwer's Fixed Point Theorem. The contribution schedule can be easily computed from the binding participation and incentive constraints. In the online [Appendix B](#) we show that the contributions must be non-negative. $\square$

Proof of Proposition 2. When $\gamma = 0$, the first-order conditions of the type- i informed lobby problem, computed in [Lemma 2](#), simplify to

$$\theta_i - \lambda [b(p_{il} - p^e) - d(\mathbf{p}_{li} - p^e)] + \frac{z}{1-z} \lambda d(\mathbf{p}_{li} - \mathbf{p}_{hi}) = 0 \quad \text{and} \quad (\text{A.13})$$

$$\theta_i - \lambda [b(p_{ih} - p^e) - d(\mathbf{p}_{hi} - p^e)] = 0. \quad (\text{A.14})$$

Eqs. (A.13) and (A.14), for lobbies 1 and 2 and all realizations of lobbies' types, constitute a system of linear equations. From [Theorem 1](#) we know that a solution to this problem exists. It remains to compute the policies. Using symmetry, it is straightforward to compute the equilibrium policies from this system of linear equations.

After some algebra, we get the expressions for the equilibrium policies presented in the statement of [Proposition 2](#), where

$$\Psi = \frac{zbd\Delta\theta_h}{\lambda((1-z)b^2 - d^2)(b^2 - d^2)}.$$

The equilibrium contributions are given by (A.3) and (A.4) calculated at these equilibrium policies.

Notice that the equilibrium policies are increasing in the lobby's own type since

$$p_{hk}^S - p_{lk}^S = \frac{b\Delta\theta_h}{\lambda(b^2 - d^2)} + \frac{zbd^2\Delta\theta_h}{\lambda((1-z)b^2 - d^2)(b^2 - d^2)} > 0, \quad \text{for all } k.$$

Thus, the SC is verified in equilibrium.

The policy maker's rents increase with the lobbies' types in equilibrium. This is so because the equilibrium contributions are computed from the policy maker's constraints. In state (θ_l, θ_l) , constraint (IRP_{ll}) is binding, and the policy maker receives his reserve utility. Since constraint (IRP_{hl}) is binding, the policy maker also receives his reserve utility in states (θ_h, θ_l) and (θ_l, θ_h) . Since constraint (ICP_{hh}) is binding, the policy maker receives positive rent when both lobbies are high-type. Therefore, the policy maker's rent is increasing in lobbies' types. So, we have that

$$U(\theta_i, p_{ih}, C_{ih}, \theta_h, \mathbf{p}_{hi}, C_{hi}) - U(\theta_i, p^e, 0, \theta_h, p^e, 0) \geq U(\theta_i, p_{il}, C_{il}, \theta_l, \mathbf{p}_{li}, C_{li}) - U(\theta_i, p^e, 0, \theta_l, p^e, 0).$$

This can be written as

$$\begin{aligned} C_{il} + C_{hi} + \lambda [W(\theta_i, p_{il}, \theta_h, \mathbf{p}_{hi}) - W(\theta_i, p^e, \theta_h, p^e)] \\ \geq C_{il} + C_{li} + \lambda [W(\theta_i, p_{il}, \theta_l, \mathbf{p}_{li}) - W(\theta_i, p^e, \theta_l, p^e)]. \end{aligned} \quad (\text{A.15})$$

¹⁸ Notice that if we take $a > 0$ sufficiently high, the truthful equilibrium policies will be in $K \times K$. Since the equilibrium policies are below the truthful policies, the equilibrium policies will never lead to zero demand. On the other hand, by the fact that contributions are positive in equilibrium and [Assumption 2](#), equilibrium policies will never be zero.

From constraint (ICP_{ih}) , we have that

$$\begin{aligned} C_{ih} + \mathbf{C}_{hi} + \lambda[W(\theta_i, p_{ih}, \theta_h, \mathbf{p}_{hi}) - W(\theta_i, p^e, \theta_h, p^e)] \\ \geq C_{il} + \mathbf{C}_{hi} + \lambda[W(\theta_i, p_{il}, \theta_h, \mathbf{p}_{hi}) - W(\theta_i, p^e, \theta_h, p^e)], \end{aligned} \quad (ICP_{ih})$$

and, from constraint (IRP_{il}) , we have that

$$C_{il} + \mathbf{C}_{li} + \lambda[W(\theta_i, p_{il}, \theta_l, \mathbf{p}_{li}) - W(\theta_i, p^e, \theta_l, p^e)] \geq 0. \quad (IRP_{il})$$

From (A.15), (ICP_{ih}) and (IRP_{il}) , we have that

$$C_{ih} + \mathbf{C}_{hi} + \lambda[W(\theta_i, p_{ih}, \theta_h, \mathbf{p}_{hi}) - W(\theta_i, p^e, \theta_h, p^e)] \geq 0.$$

Therefore, the fact that the policy maker's rent is increasing and condition (A.15) together with the binding constraints (IRP_{il}) and (ICP_{ih}) all ensure that constraint (IRP_{ih}) is satisfied in equilibrium.

Since constraint (ICP_{ih}) is binding, condition (A.1) holds with equality which allows us to compute the contribution C_{ih} . Since (A.1) is binding and policies increase with lobbies' types, constraint (A.2) holds with inequality, and constraint (ICP_{il}) is not binding. Therefore, the postulated binding constraints hold in equilibrium. $\square$

Proof of Corollary 1. We must compare the expected welfare at the screening equilibrium with the welfare at the truthful equilibrium. Since preferences are quasi-linear and concave, the distribution of contributions does not affect the size of society's welfare. Therefore, we only need to compare these expected welfare functions evaluated at the equilibrium policies, that is, to compare $W(\theta_i, \bar{p}_{ik}, \theta_k, \bar{p}_{ki})$ with $W(\theta_i, p_{ik}^S, \theta_k, p_{ki}^S)$. In what follows we will do it state by state.

In state (θ_h, θ_h) , policies are the same for both truthful and screening equilibria. Therefore, the welfare of the society is also the same. The policies in the other states are such that

$$\begin{aligned} p_{hl}^S &= \frac{\theta_h b \left(\frac{(1-z)b-d}{b-d} \right) + \theta_l d}{\lambda((1-z)b^2 - d^2)} + p^e < \bar{p}_{hl}, \\ p_{lh}^S &= \frac{\theta_l b(1-z) + \theta_h d \left(\frac{(1-z)b-d}{b-d} \right)}{\lambda((1-z)b^2 - d^2)} + p^e < \bar{p}_{lh} \text{ and} \\ p_{ll}^S &= \frac{\theta_l(b(1-z) + d) - \theta_h d \left(\frac{zb}{b-d} \right)}{\lambda((1-z)b^2 - d^2)} + p^e < \bar{p}_{ll}. \end{aligned}$$

The maximum welfare for the society is given by the free trade equilibrium policy p^e . In turn, this implies that $\partial W / \partial p^n < 0$, for all $p > p^e$. In states (θ_h, θ_l) and (θ_l, θ_h) , policies of the screening equilibrium are such that

$$\bar{p}_{hl} > p_{hl}^S > p^e \quad \text{and} \quad \bar{p}_{lh} > p_{lh}^S > p^e,$$

i.e., they are below the truthful policies and above international prices (because $(1-z)b > d$). Therefore, the welfare at the screening equilibrium is higher than at the truthful equilibrium in these states.

In state (θ_l, θ_l) , equilibrium policies are also below the truthful policies, but they can fall below international prices as well. Hence, to show that the welfare of the political game in this state is greater than the welfare of the truthful equilibrium, we must compare the welfare of the two equilibria directly from the expressions, that is, we must show that

$$W(\theta_l, \bar{p}_{ll}, \theta_l, \bar{p}_{ll}) \leq W(\theta_l, p_{ll}^S, \theta_l, p_{ll}^S).$$

Given our functional forms, the last inequality is equivalent to

$$(\bar{p}_{ll} - p^e)^2 - (p_{ll}^S - p^e)^2 \geq 0.$$

Since $p_{ll}^S < \bar{p}_{ll}$, we must have that

$$(\bar{p}_{ll} - p^e)^2 \geq (p_{ll}^S - p^e)^2.$$

If $p_{ll}^S > p^e$, the above inequality holds. However, it is possible that $p_{ll}^S < p^e$, in which case the above inequality becomes $\bar{p}_{ll} - p^e \geq p^e - p_{ll}^S$. Replacing the policies by their closed form solutions, this last inequality becomes

$$\frac{\theta_l(b+d)}{\lambda(b^2 - d^2)} \geq -\frac{\theta_l(b+d)}{\lambda(b^2 - d^2)} + \frac{zbd(b+d)\Delta\theta_h}{\lambda((1-z)b^2 - d^2)(b^2 - d^2)}.$$

After some algebra, the last expression simplifies to

$$\frac{(1-z)b^2 - d^2}{zbd} > \frac{\Delta\theta_h}{2\theta_l},$$

which holds by [Assumption 2](#). Therefore, in state (θ_l, θ_l) the welfare at the equilibrium is greater than at the truthful equilibrium, which concludes the proof. $\square$

Proof of Proposition 3. In this proof we solve the informed lobby problem for the case where $d = 0$ and $\gamma = 1$. For these parameters, the system of best responses derived from problem (6) is simplified. The first simplification is that constraints (ICP_{ik}) are equivalent to constraints (IRP_{ik}) . The last ones are binding for all realizations of the lobbies' types. Also, for $d = 0$, the policy maker's preferences are separable in the arguments (see Eq. (1)). As a consequence, the lobby's policy is invariant to her rival's type. So, we drop the second subscript index. Hence, the constraint (IC_h) may be binding.

These simplifications allow us to eliminate the multiplier μ_h from the first-order conditions derived in [Lemma 2](#). The resulting system is given by

$$\theta_h p_h - \lambda(b + \theta_h)(p_h - p^e) = 0, \quad (\text{A.16})$$

$$\theta_l p_l - \lambda(b + \theta_l)(p_l - p^e) \geq 0, \quad (\text{A.17})$$

$$V(\theta_h, p_h, C_h) - V(\theta_h, p_l, C_l) \geq 0 \quad \text{and} \quad (\text{IC}_h)$$

$$[\theta_l p_l - \lambda(b + \theta_l)(p_l - p^e)][V(\theta_h, p_h, C_h) - V(\theta_h, p_l, C_l)] = 0, \quad (\text{A.18})$$

and the contributions are computed from the binding constraints (IRP_l) .

The first two equations of the above system are the informed lobby problem's first-order conditions. The low-type lobby's first-order condition is an inequality because if she has to separate herself, constraint (A.17) may not be binding. The third equation is the high-type lobby's incentive compatibility constraint (IC_h) . The fourth equation states that either the low-type's first-order condition or the constraint (IC_h) are binding. That is, separation may or may not be costly.

Now, we must find the policies that solve Eqs. (A.16)–(A.18). From (A.16), we have that

$$\bar{p}_h = \frac{\lambda(b + \theta_h)}{\lambda(b + \theta_h) - \theta_h} p^e,$$

which is the high-type lobby's truthful policy.

If the low-type lobby's first-order condition is binding, we get the low-type lobby's truthful policy

$$\bar{p}_l = \frac{\lambda(b + \theta_l)}{\lambda(b + \theta_l) - \theta_l} p^e.$$

On the other hand, if constraint (IC_h) is binding, we have

$$\frac{1}{2}\theta_h p_h^2 - \lambda \frac{(b + \theta_h)}{2}(p_h - p^e)^2 = \frac{1}{2}\theta_l p_l^2 - \lambda \frac{(b + \theta_l)}{2}(p_l - p^e)^2.$$

Replacing p_h by the expression of $\bar{p}_h$ gives the following solution, after some algebra:

$$\check{p}_l = \frac{\lambda(b + \theta_l) - \theta_h \sqrt{\frac{\lambda \Delta \theta_h}{\lambda(b + \theta_h) - \theta_h}}}{\lambda(b + \theta_l) - \theta_h} p^e.$$

The issue now is to define when the constraint (IC_h) is binding. Suppose that $\check{p}_l \geq \bar{p}_l$, i.e., the policy which ensures separation is greater than the truthful policy. Then,

$$V(\theta_h, \bar{p}_h, C_h(\bar{p}_h)) = V(\theta_h, \check{p}_l, C_l(\check{p}_l)) \geq V(\theta_h, \bar{p}_l, \bar{C}_l(\bar{p}_l)), \quad (\text{A.19})$$

where contributions $C_i(p)$ computed from (IRP_i) are functions of the policies. Moreover, the equality holds by the definition of $\check{p}_l$ and the inequality holds because $V(\theta_h, p_l, C_l(p_l))$ is increasing in p_l , for all $p_l \in [0, \frac{\lambda(b + \theta_l)}{\lambda(b + \theta_l) - \theta_h}]$.¹⁹

Therefore, $\bar{p}_h$ solves the first-order condition of the high type and $\bar{p}_l$ solves the first-order condition of the low type (with equality). From (A.19), the incentive compatibility constraint (IC_h) is not binding at $p_l = \bar{p}_l$. Eq. (A.18) holds trivially since the first-order condition of the low type holds with equality. Hence, $\bar{p}_h$ and $\bar{p}_l$ solve the system of Eqs. (A.16)–(A.18) and are the equilibrium policies, i.e., $\bar{p}_h = p_h^*$ and $\bar{p}_l = p_l^*$. In particular, $\bar{p}_l$ and $\bar{p}_h$ solve Eq. (A.17) and, from (A.19), we have that constraint (IC_h) is not binding. Therefore, separation is achieved without any signaling cost.

¹⁹ This results from the fact that $p_l = \frac{\lambda(b + \theta_l)}{\lambda(b + \theta_l) - \theta_h}$ maximizes $V(\theta_h, p_l, C_l(p_l))$ subject to (IRP_l) .

On the other hand, if $\check{p}_l < \bar{p}_l$, Eq. (A.17) implies that $\theta_l \check{p}_l - \lambda(b + \theta_l)(\check{p}_l - p^e) > 0$. Additionally, constraint (IC_h) is binding by the construction of $\check{p}_l$. Since constraint (IC_h) is binding, (A.18) trivially holds. Therefore, $\bar{p}_h$ and $\check{p}_l$ solve the system of Eqs. (A.16)–(A.18), i.e., $\bar{p}_h = p_h^*$ and $\check{p}_l = p_l^*$ are the equilibrium policies. Moreover, since constraint (IC_h) is binding, separation is costly. Notice also that in both cases the expression for the high-type policy remains the same.

The following threshold which indicates whether $\check{p}_l$ or $\bar{p}_l$ are the equilibrium policy for the low type lobby:

$$\check{p}_l < \bar{p}_l \iff \bar{p}_h - p^e > \frac{\Delta\theta_h}{\lambda\theta_h} \bar{p}_l^2.$$

In such case, constraint (IC_h) is binding and $\check{p}_l$ is the equilibrium policy for the low-type lobby. Otherwise, we have that

$$\check{p}_l \geq \bar{p}_l \iff \bar{p}_h - p^e \leq \frac{\Delta\theta_h}{\lambda\theta_h} \bar{p}_l^2,$$

and then $\bar{p}_l$ is the equilibrium policy for the low-type lobby. This completes the proof.

Finally, let us show that the equilibrium policies are greater than the international price p^e .

(i) If $\bar{p}_h - p^e \leq (\Delta\theta_h/\lambda\theta_h)\bar{p}_l^2$, the equilibrium policies are the truthful policies. From Assumption 2 it is easy to check that they are indeed greater than p^e .

(ii) On the other hand, if $\bar{p}_h - p^e > (\Delta\theta_h/\lambda\theta_h)\bar{p}_l^2$, then $p_l^* = \check{p}_l < \bar{p}_l$. Suppose by contradiction that $\check{p}_l < p^e$. Then,

$$V(\theta_h, \check{p}_l, C_l(\check{p}_l)) < V(\theta_h, p^e, 0),$$

because contributions are non-negative (see the online Appendix B) and $C(p^e) = 0$. Hence, we have that

$$V(\theta_h, p^e, 0) > V(\theta_h, \check{p}_l, C_l(\check{p}_l)) = V(\theta_h, \bar{p}_h, C_h(\bar{p}_h)),$$

where the second equality follows from the definition of $\check{p}_l$. This is a contradiction because, from Assumption 2, in a truthful equilibrium the lobbies get rents and positive policies. This gives a greater utility than her reserve utility. Therefore, both equilibrium policies are greater than the international price p^e . $\square$

Proof of Proposition 4. In this proof we characterize the equilibrium of the informed lobby problem for the case where $d > 0$ and $\gamma = 1$. The first-order conditions of the high- and low-type informed-lobby problems computed in Lemma 2 are given by

$$\theta_l p_{ll} - \lambda(b + \theta_l)(p_{ll} - p^e) + \lambda d(\mathbf{p}_{ll} - p^e) - \frac{\mu \Delta\theta_h}{1 - \mu} p_{ll} + \lambda d \frac{z}{1 - z} (\mathbf{p}_{ll} - \mathbf{p}_{hl}) = 0, \quad (\text{A.20})$$

$$\theta_l p_{lh} - \lambda(b + \theta_l)(p_{lh} - p^e) + \lambda d(\mathbf{p}_{hl} - p^e) - \frac{\mu \Delta\theta_h}{1 - \mu} p_{lh} = 0, \quad (\text{A.21})$$

$$\theta_h p_{hl} - \lambda(b + \theta_h)(p_{hl} - p^e) + \lambda d(\mathbf{p}_{lh} - p^e) + \lambda d \frac{z}{1 - z} (\mathbf{p}_{hl} - \mathbf{p}_{hh}) = 0 \quad (\text{A.22})$$

and

$$\theta_h p_{hh} - \lambda(b + \theta_h)(p_{hh} - p^e) + \lambda d(\mathbf{p}_{hh} - p^e) = 0. \quad (\text{A.23})$$

Notice also that $\mu = \mu_h$ since, from Lemma 3, only constraint (IC_h) can be binding.

Eqs. (A.20)–(A.23) together with constraint (IC_h) , for both lobbies, constitute a system of linear equations that can be explicitly solved. From Theorem 1 we know the equilibrium always exists.

Using symmetry, the equilibrium policies are not difficult to compute for a given value of μ . However, since the expressions are not as neat as in the pure screening case, we only present the expression of p_{lh}^* for a given value of μ :

$$p_{lh}^* - p^e = \frac{(1 - z)(\theta_l - \frac{\mu \Delta\theta_h}{1 - \mu})(\lambda(b + \theta_h) - \theta_h) + \lambda d \theta_h \frac{(1 - z)(\lambda(b + \theta_h) - \theta_h) - \lambda d}{\lambda(b + \theta_h) - \theta_h - \lambda d}}{(1 - z)(\lambda(b + \theta_h) - \theta_h)(\lambda(b + \theta_l) - \theta_l + \frac{\mu \Delta\theta_h}{1 - \mu}) - (\lambda d)^2} p^e < \bar{p}_{lh}.$$

It is hard to compute the explicit value of μ when constraint (IC_h) is binding. Yet, μ is such that p_{ll}^* and p_{lh}^* satisfy

$$\frac{\theta_l p_{ll}^* + \lambda \frac{\partial W}{\partial p_{ll}}(\theta_l, p_{ll}^*, \theta_l, \mathbf{p}_{ll}) + \frac{z}{1 - z} \lambda d[\mathbf{p}_{ll} - \mathbf{p}_{hl}]}{\theta_h p_{ll}^* + \lambda \frac{\partial W}{\partial p_{ll}}(\theta_l, p_{ll}^*, \theta_l, \mathbf{p}_{ll}) + \frac{z}{1 - z} \lambda d[\mathbf{p}_{ll} - \mathbf{p}_{hl}]} = \frac{\theta_l p_{lh}^* + \lambda \frac{\partial W}{\partial p_{lh}}(\theta_l, p_{lh}^*, \theta_h, \mathbf{p}_{hl})}{\theta_h p_{lh}^* + \lambda \frac{\partial W}{\partial p_{lh}}(\theta_l, p_{lh}^*, \theta_h, \mathbf{p}_{hl})},$$

and the binding constraint (IC_h) .

Moreover, it is straightforward to verify that constraints (IC_{ph}) , (IC_{lh}) , (IR_{ph}) , and (IR_{pl}) are binding, while constraint (IC_h) may or may not be binding, as in Propositions 2 and 3. All other constraints are not binding in equilibrium. $\square$

Proof of Corollary 2. This proof shows that the welfare for the equilibrium of the political game when $\gamma = 1$ (that can be found in Proposition 4) is greater than the welfare for the truthful equilibrium.

We know that the impact of policies on welfare is negative for all policies above p^e . Since the equilibrium policies are such that $\bar{p}_{ik} \geq p_{ik}^* > p^e$ (with at least one strict inequality) we have that

$$W(\theta_i, \bar{p}_{ik}, \theta_k, \bar{p}_{ki}) < W(\theta_i, p_{ik}^*, \theta_k, p_{ki}^*).$$

Therefore, the welfare at the equilibrium of the political game with informed lobbies is greater than the welfare at the truthful equilibrium. $\square$

Appendix B. Supplementary material

Supplementary material related to this article can be found online at <http://dx.doi.org/10.1016/j.geb.2014.02.010>.

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